



A role for the serotonin 2A receptor in the expansion and functioning of human transmodal cortex

Andrea I. Luppi,^{1,2,3,†} Manesh Girn,^{4,5,†} Fernando E. Rosas,^{6,7,8} Christopher Timmermann,⁶ Leor Roseman,⁶ David Erritzoe,⁶ David J. Nutt,⁶ Emmanuel A. Stamatakis,¹ R. Nathan Spreng,⁴ Lei Xing,⁹ Wieland B. Huttner⁹ and Robin L. Carhart-Harris^{5,6}

[†]These authors contributed equally to this work.

Integrating independent but converging lines of research on brain function and neurodevelopment across scales, this article proposes that serotonin 2A receptor (5-HT_{2A}R) signalling is an evolutionary and developmental driver and potent modulator of the macroscale functional organization of the human cerebral cortex. A wealth of evidence indicates that the anatomical and functional organization of the cortex follows a unimodal-to-transmodal gradient. Situated at the apex of this processing hierarchy—where it plays a central role in the integrative processes underpinning complex, human-defining cognition—the transmodal cortex has disproportionately expanded across human development and evolution. Notably, the adult human transmodal cortex is especially rich in 5-HT_{2A}R expression and recent evidence suggests that, during early brain development, 5-HT_{2A}R signalling on neural progenitor cells stimulates their proliferation—a critical process for evolutionarily-relevant cortical expansion. Drawing on multimodal neuroimaging and cross-species investigations, we argue that, by contributing to the expansion of the human cortex and being prevalent at the apex of its hierarchy in the adult brain, 5-HT_{2A}R signalling plays a major role in both human cortical expansion and functioning. Owing to its unique excitatory and downstream cellular effects, neuronal 5-HT_{2A}R agonism promotes neuroplasticity, learning and cognitive and psychological flexibility in a context-(hyper)sensitive manner with therapeutic potential. Overall, we delineate a dual role of 5-HT_{2A}R in enabling both the expansion and modulation of the human transmodal cortex.

- 1 Department of Clinical Neurosciences and Division of Anaesthesia, University of Cambridge, Cambridge, CB2 0QQ, UK
- 2 Leverhulme Centre for the Future of Intelligence, University of Cambridge, Cambridge, CB2 1SB, UK
- 3 The Alan Turing Institute, London, NW1 2DB, UK
- 4 Montreal Neurological Institute, Department of Neurology and Neurosurgery, McGill University, Montreal, H3A 2B4, Canada
- 5 Psychedelics Division—Neuroscape, Department of Neurology, University of California San Francisco, San Francisco, CA 94158, USA
- 6 Centre for Psychedelic Research, Department of Brain Sciences, Faculty of Medicine, Imperial College London, London, SW7 2AZ, UK
- 7 Data Science Institute, Imperial College London, London, SW7 2AZ, UK
- 8 Centre for Complexity Science, Imperial College London, London, SW7 2AZ, UK
- 9 Max Planck Institute of Molecular Cell Biology and Genetics, Dresden, 01307, Germany

Correspondence to: Manesh Girm

University of California San Francisco, Sandler Neurosciences Center, 675 Nelson Rising Lane, San Francisco, CA 94158, USA

E-mail: manesh.girm@ucsf.edu

Keywords: psychedelics; transmodal cortex; serotonin 2A receptor; cortical expansion; neurodevelopment; cortical functional organization

Introduction

Neuroscience has long sought to understand how the size and complexity of the human cerebral cortex relates to the remarkable cognitive capacities of our species. This line of inquiry has increasingly highlighted the central role of the human transmodal association cortex: the set of limbic, paralimbic and heteromodal regions whose activity and connectivity reflect the higher-order integration of inputs from multiple modalities.^{1,2} Contrasted with the rest of cortex, human transmodal association cortex has undergone a remarkable and disproportionate degree of expansion relative to non-human primates^{1,3–6}—an expansion that is also mirrored by protracted ontogenetic development, with developmental trajectories extending into the second decade of life.³ In addition, a multimodal body of research has increasingly identified a set of anatomical, genetic, molecular, physiological and functional features that set transmodal cortex apart from unimodal cortex and which are thought to enable the functional complexity necessary for the emergence of human cognitive, socio-emotional and cultural functioning.^{1,5,7–11} This research has revealed that a continuous gradient of variation from unimodal to transmodal cortex may constitute the primary macroscale organizational scheme of the cortex.¹ Such findings are consistent with seminal and highly influential work which, on the basis of anatomical characteristics derived from tract-tracing and histology, identified a ‘sensory-fugal’ hierarchical axis spanning the cortical mantle, moving from modality-specific (primary and unimodal) sensory processing to multimodal integration, to higher-order integrative processing in transmodal cortices.²

Collectively, research to date suggests that the transmodal cortices represent the apex of the macroscale cortical hierarchy—from specialized, concrete unimodal processing to integrative, abstract transmodal processing—and play a central role in orchestrating human cognitive and behavioural capacities. Consistent with this, task-based functional neuroimaging investigations have implicated the transmodal cortices in a range of ‘high-level’ cognitive processes including attention and executive cognition, episodic and semantic memory, social cognition and narrative comprehension.^{12,13} A large body of work has also implicated disruptions of transmodal cortex structure and/or function in a variety of neurological and psychiatric illnesses.^{1,14,15}

Interestingly, a growing body of evidence suggests that acute pharmacological modulation of the transmodal cortices may have therapeutic potential.^{16,17} This work consists of investigations with serotonergic psychedelic drugs such as psilocybin and lysergic acid diethylamide (LSD), compounds that elicit their primary effects via partial agonism at the 5-HT_{2A} receptor (5-HT_{2A}R)—an excitatory receptor that is most densely expressed in transmodal cortices.¹⁸ Several preliminary clinical trials have found that 5-HT_{2A}R agonist psychedelic drugs, when combined with supportive psychotherapy, can induce long-lasting symptom reductions

following only one to three drug sessions.¹⁹ Evidence is presently strongest for depression^{20–24} but suggestive results have also been found for end-of-life distress in terminal patients,^{25,26} tobacco addiction²⁷ and alcoholism.²⁸ Research with psychedelics has also specifically implicated 5-HT_{2A}R in plasticity- and flexibility-promoting processes, at structural, functional and behavioural levels.^{29–34} This suggests a unique ability for this class of compounds to modulate or transiently upregulate the properties characteristic of transmodal cortex—with therapeutic relevance.

In addition, recent findings have also begun providing support for a potential role of 5-HT_{2A}R signalling in mammalian cortical expansion—both phylogenetically and ontogenetically—especially in relation to the disproportionate expansion of human transmodal cortex. Notably, a recent multi-species study found that 5-HT_{2A} signalling in early developing cortical tissue significantly promoted the proliferation of basal progenitors that putatively underlie the evolutionary expansion of the human cortex.³⁵ This pro-proliferative role in basal progenitors appears to be unique amongst the neurotransmitters and neurotransmitter receptors³⁶ and is consistent with a large body of work implicating serotonin (5-HT) in a variety of critical neurodevelopmental processes.^{37,38} In this regard, it is also intriguing to highlight that high resolution *in vivo* PET molecular imaging of 5-HT receptor distributions in the adult human brain has revealed that the spatial topography of 5-HT_{2A}R densities strongly resembles the unimodal-transmodal cortical gradient, with highest densities in transmodal cortex.¹⁸ (We note that primary visual cortex constitutes a notable exception to this pattern, given that it is situated at the opposite end of the cortical hierarchy from transmodal association cortex but is also rich in 5-HT_{2A}R expression. We return to this point in later sections.)

Overall, there is converging evidence that: (i) 5-HT_{2A}Rs are most densely expressed in the disproportionately expanded transmodal cortex of the human brain; (ii) 5-HT_{2A}Rs are core contributors to both the ontogenetic and phylogenetic expansion of transmodal cortex; and (iii) 5-HT_{2A}R agonism, particularly via serotonergic psychedelics, can potentially modulate the functioning of transmodal cortex, thereby engaging neural and behavioural plasticity in the adult brain. In the present article, we focus on 5-HT_{2A}R, bringing together these independent but complementary lines of research to provide an integrative account of the role of 5-HT_{2A}R signalling in shaping the developmental expansion and adult functioning of the human transmodal cortex. We argue that thanks to the role that they play in the expansion of transmodal cortex—the apex of the human cortical hierarchy—5-HT_{2A}Rs may become especially well positioned to subsequently modulate the adult functioning of transmodal cortex. We highlight how this is supported by nascent research on 5-HT_{2A}R agonist serotonergic psychedelic drugs, which have been found to induce complex and wide-ranging subjective effects, alongside a variety of therapeutically-relevant acute and post-acute structural, functional and behavioural changes.

Transmodal association cortex: the centrepiece of human cognitive architecture

Hierarchical organization of the human cerebral cortex

Humans' 'success' as a remarkably populous species is unquestionably linked to our ability to engage in complex cognition and cognitive neuroscience has revealed that our species' high-order cognitive faculties are fundamentally dependent on the outermost component of the human brain: the cerebral cortex. The human cerebral cortex is an exceptionally complex organ, with marked regional anatomical heterogeneity. Investigations of cortical variation in cytoarchitectonics and connectional anatomy^{39–42} have delineated a continuous 'sensory-fugal' hierarchy from primary sensory and unimodal cortices to transmodal association cortices.^{2,41} According to this scheme, each end of the hierarchy processes inputs of a different nature: whereas unimodal cortex only responds to stimuli pertaining to one specific modality (e.g. vision or audition), transmodal cortex is situated at the convergence of multiple sensory streams² and this organization outlines a progression from domain-specific sensory processing to integrative domain-general abstract processing.

Remarkably, a rapidly growing body of convergent multimodal evidence suggests that the unimodal-transmodal hierarchy represents an 'archetypal axis' that delineates principal dimensions of several axes of functional, structural, cellular and molecular variation across the cortex (Fig. 1).^{1,43} This work has found that, relative to unimodal cortex, regions closer to the transmodal (and especially heteromodal) apex of the axis are characterized by lower neuron density,⁴⁴ a predominance of infragranular (feedback) efferent connections,¹⁰ lower laminar differentiation,⁴⁵ lower intracortical myelination^{7,46,47} and greater aerobic glycolysis,⁴⁸ increased excitability and greater density of large and dendritically complex pyramidal cells,^{44,49–51} greater cortical thickness^{3,52,53} and lower structure-function coupling.^{54–56} This macroscale unimodal to transmodal hierarchy based on neuroanatomical considerations is also recapitulated by the principal axis of variation in intrinsic cortical functional connectivity from functional MRI.^{8,43} In addition, functional connectivity research has found that cortical signals propagate in a sensory-fugal fashion from specialized and modular sensory processing in unimodal cortex, to distributed and integrative processing in transmodal cortices.^{54,57} The convergence of these characteristics is thought to confer the unique functional properties of transmodal cortex, which afford complex human behaviour and cognition, as detailed in the following sections.

Transmodal association cortices orchestrate higher cognitive function

Transmodal association cortices represent the point of convergence for diverse modality-specific inputs.^{2,58,59} This anatomical insight is reflected at the functional level. At the lower, sensorimotor end of the hierarchy, localized electrical stimulation induces modality-specific sensations—whereas the elicited experiences become richer and multimodal upon stimulation of transmodal associations cortices.⁶⁰ (The human transmodal cortex as neuroanatomically defined in early work is divided into the cytoarchitectonically and connectionally distinct heteromodal, limbic and paralimbic transmodal cortices.² In the present context we focus primarily on

heteromodal transmodal cortex, which represents the integrative apex of transmodal cortex itself.)

Across a variety of task paradigms, functional MRI (fMRI) has revealed that primary cortices exhibit preferential involvement with modality-specific tasks and processing, such as motor control and visual/auditory/somatosensory stimulation.^{8,61,62} In contrast, transmodal (and in particular, heteromodal) association cortices show relatively greater engagement during complex cognition. Even at rest, canonical 'intrinsic networks' are consistently observed across participants, which closely resemble activation patterns observed with task-based analyses.^{63,64} This work has revealed that the transmodal cortex can be subdivided into at least two distinct intrinsic networks, typically referred to as the 'frontoparietal control network' and the 'default mode network' (DMN). The frontoparietal control network mainly comprises lateral prefrontal and parietal cortices, and is recruited during engagement with cognitively demanding tasks, irrespective of modality.^{13,65} The DMN's key components include the posterior cingulate and precuneus, medial prefrontal cortex and (bilateral) inferior parietal cortices.⁶⁶ Although also capable of supporting externally-directed cognition in tasks that require or are facilitated by past knowledge,^{67–70} the DMN is especially involved in abstract cognitive operations that rely upon perceptually decoupled mnemonic information and transcend the here-and-now.^{12,71,72}

Taken together, data-driven functional investigations, as well as causal evidence from brain lesions and stimulation, converge on the conclusion that cognitive sub-specialization is present within the brain and that transmodal cortex is particularly involved in domain-general and complex forms of cognition that are most characteristic of humans.

Flexibility as key feature of transmodal association cortex

Having established the empirical relevance of transmodal association cortex for high-order human cognition, we are left with a central question: Why is transmodal cortex well suited to orchestrate high-order cognitive functions? We believe the answer lies in the exceptional 'functional flexibility' of the human transmodal cortex (and especially, heteromodal cortex)—an 'umbrella' construct or property that can be understood in multiple complementary ways (Fig. 2).

First, the flexibility of regions at the top of the cortical hierarchy is evident in terms of diversity, in terms of several characteristics: they exhibit the widest dynamic range of spontaneous temporal fluctuations⁷⁴; and diverse (highly variable) patterns of intrinsic functional connectivity⁷⁷; they adaptively shift their connectivity patterns in response to task demands,⁷⁶ while balancing flexibility and specialization⁷⁸; and they exhibit the greatest diversity of neurotransmitter receptors across layers, as quantified from post-mortem autoradiography.¹⁰ Taken together, this evidence helps to explain how the transmodal association cortices can produce highly adaptive and flexible responses.

Second, the workings of the apex of the hierarchy are flexible in terms of their relative independence from the dictates of anatomy. Sensory cortices are strongly tethered to input from the sensory organs (relayed via the thalamus) but the same is not true of the transmodal cortices. Relatedly, functional and structural connectivity are increasingly decoupled along the cortical hierarchy.^{55,56,73} More broadly, transmodal association cortices are developmentally constrained by the brain's myeloarchitecture and molecular and transcriptomic gradients to a lesser extent than are the unimodal

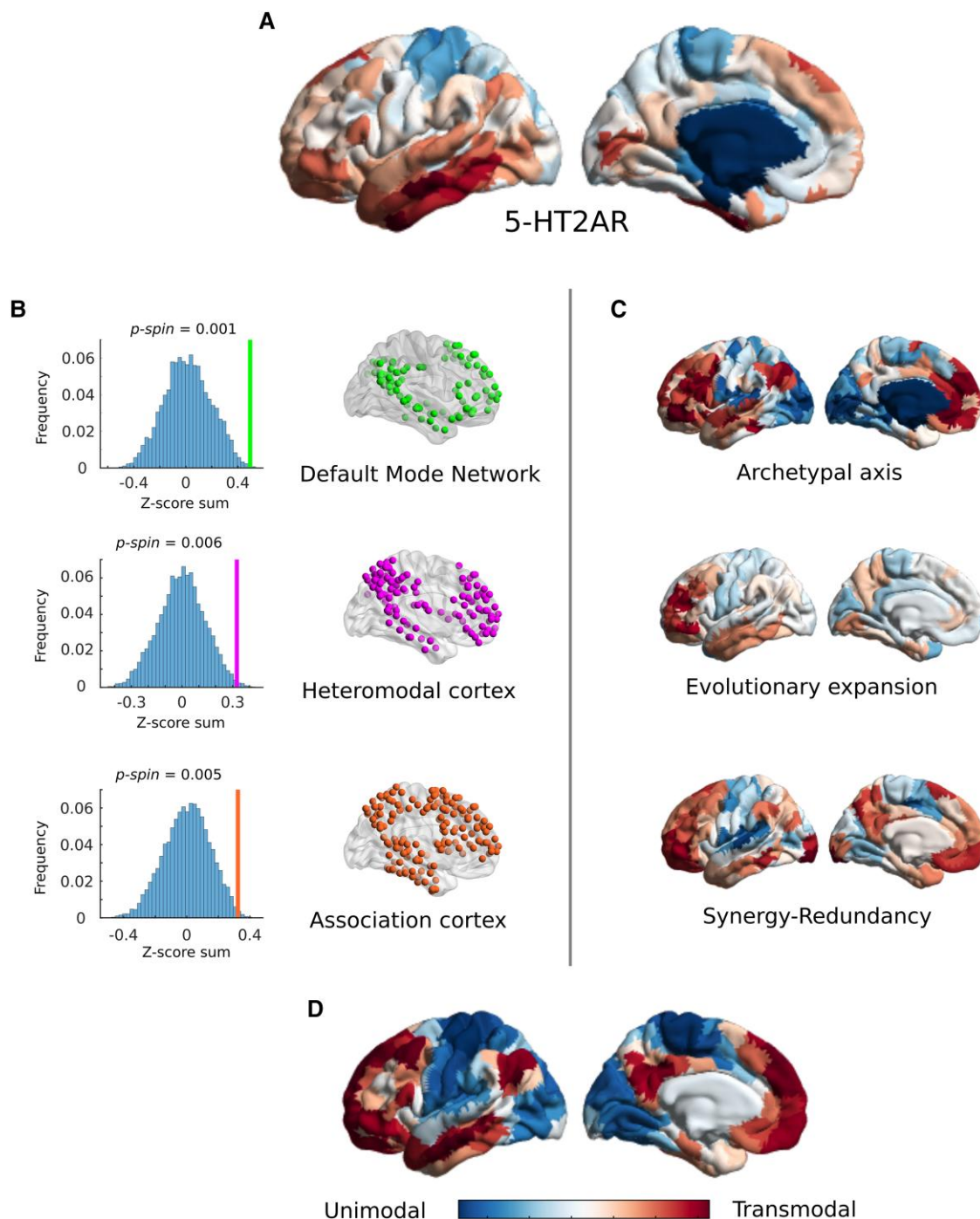


Figure 1 Hierarchical distribution of 5-HT2ARs in the human cortex. (A) A recent high resolution map of the regional availability of 5-HT2ARs in the human brain obtained from *in vivo* PET imaging.¹⁸ (B) We show that the cortical 5-HT2AR distribution is significantly enriched at the apex of the cortical hierarchy, whether defined in functional terms (default mode network), or anatomical feed-forward projections (Mesulam's heteromodal cortex, which is part of transmodal cortex); or cytoarchitectonics (association cortex from Von Economo's classification). In each case, significance ('p-spin') is assessed against a null distribution with preserved spatial autocorrelation, with a coloured vertical bar indicating the empirically observed value.¹¹⁴ (C) We also show that serotonin 2A receptor densities in the human cortex are spatially aligned with the regional pattern of cortical expansion with respect chimpanzees (*P. troglodytes*), the species closest to *Homo sapiens* in evolutionary terms⁴; a recently defined 'archetypal axis' of cortical organization, obtained by combining 10 distinct gradients of cortical variation defined from functional, structural, cytoarchitectonic, myeloarchitectonic, genetic and metabolic evidence¹; and a gradient from redundancy-dominated to synergistic information processing, based on functional neuroimaging.¹¹⁰ (D) Functional characterization of the unimodal-transmodal gradient, based on Margulies et al.⁸

cortices.⁴⁷ Indeed, molecular and transcriptomic gradients have, themselves, been shown to delineate hierarchies related to both anatomy and cognition.^{9,11,79} Finally, it is also worth noting that,

although most pronounced in humans, transmodal cortices have also been found to be less structurally constrained relative to unimodal cortices in macaques.⁸⁰

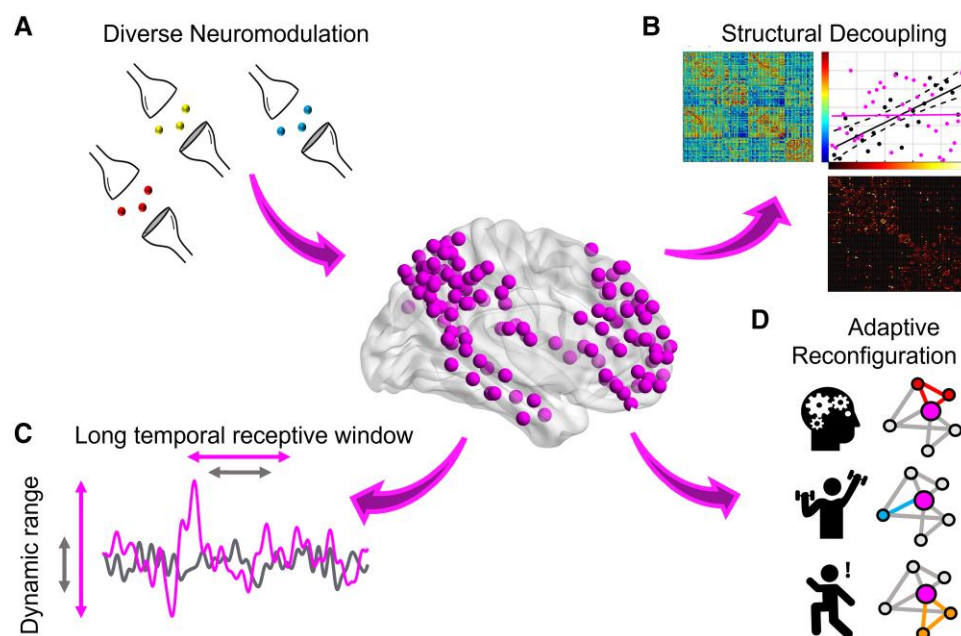


Figure 2 Flexibility of transmodal association cortex. Transmodal association cortex is flexible across multiple dimensions. (A) It exhibits the most diverse patterns of neurotransmitter receptors.¹⁰ (B) Seed-based patterns of functional connectivity centred in transmodal cortex are relatively decoupled from the underlying patterns of macroscale structural connections^{55,56,73}; purple elements of the scatter-plot indicate correlation between entries of the functional connectivity matrix (y-axis) and structural connectivity matrix (x-axis) for a region in transmodal cortex; black elements reflect the structure-function correlation for a region in unimodal cortex. (C) Activity in transmodal cortices exhibits relatively long windows of temporal integration and a wide dynamic range.^{74,75} (D) Transmodal cortices exhibit varying connectivity in response to different task demands.⁷⁶

Third, the apex of the hierarchy is flexible in terms of its independence from immediate sensorimotor contingencies. This is reflected in differences in the temporal characteristics of regional activity. The transmodal association cortices are characterized by temporally extended ‘receptive windows’, enabling them to reflect and bring together information from events taking place across greater periods of time.^{75,81–84} For example, research with naturalistic movie-viewing has found that, while unimodal cortices track second-by-second changes in sensory information, transmodal cortices integrate scene/event-related information over multiple seconds to minutes to support abstract, multimodal interpretational processes.^{85,86} This corresponds to slower intrinsic dynamics, which have been observed to arise from structural considerations in regions of the brain’s densely connected ‘structural core’⁸⁷ but may also be related to higher density of NMDA NR2B, which prolong excitatory synaptic activity.¹ As such, the spatial unimodal-to-transmodal hierarchy can be recapitulated by a temporal hierarchy of intrinsic timescales.⁷⁵ The independence of transmodal cortices from immediate sensorimotor contingencies is also reflected in their spatial embedding: regions within transmodal association cortex are spatially the most distant from sensory and motor regions along the cortical surface^{6,8} and are functionally the most distant, as evidenced by their occupation of the opposite end of the principal hierarchical gradient of functional connectivity similarity.⁸ This is consistent with the DMN’s role in going beyond the here-and-now by bringing perceptually decoupled mnemonic information to bear on ongoing experience and task demands.^{12,67,69,70,88} It is also consistent with the executive control network’s role in inhibiting prepotent responses evoked by immediate circumstances and the selection of alternative actions.^{65,89,90} Thus, evidence indicates that transmodal cortices are less constrained by the here-and-now in terms of their functioning. In analogy with today’s deep-learning architectures, in virtue of its

location at the apex of the cortical processing hierarchy, the transmodal cortex may be thought of as the brain’s ‘deepest layer’, providing the opportunity for the behavioural outputs to be informed by complex, situation- and task-specific combinations of inputs, rather than a limited range of predetermined, hard-wired input-output mappings.^{12,88,91}

Collectively, the above-discussed body of work suggests that the ability for the transmodal association cortex to support the adaptive, flexible and integrative processes underlying complex human cognition can be attributed to its functional and connectational diversity, relative independence from the dictates of anatomy and relative freedom from incoming sensory information. Next, we describe research which suggests that this functional flexibility is itself scaffolded upon high anatomical plasticity—what we refer to as ‘meta-flexibility’.

Meta-flexibility: plasticity of transmodal association cortex

In addition to these functional definitions of flexibility, the apex of the cortical hierarchy is also flexible in another important sense: it has an exceptionally high capacity for undergoing neuroplastic change over the lifespan. In other words, the functional characteristics described above are themselves liable to change in a flexible manner. In addition, the transmodal cortex exhibits the lowest levels of intracortical myelination, as measured non-invasively by the ratio of T₁-weighted to T₂-weighted MRI.^{7,47} This is relevant because evidence indicates that after closure of the ‘critical period’ of brain development (i.e. a temporally restricted period of heightened sensitivity to environmental factors that is relevant for neural maturation), myelin suppresses excitatory synaptic plasticity: both by constituting a physical barrier to the emergence of new neurites, and by means of myelin-associated ‘nogo’ receptor

(NgR1) signalling, which has inhibitory effects.^{92,93} Thus, by being comparatively low in intracortical myelination, the top of the cortical hierarchy has greater potential for synaptic plasticity.

Transmodal association cortices are also characterized by metabolic differences from other cortices, which are relevant for their capacity for plasticity. Specifically, they exhibit the highest rates of aerobic glycolysis, which is a metabolic cycle whereby energy is extracted from glucose through non-oxidative metabolism rather than CO₂-producing oxidative metabolism.⁴⁸ The unique products of aerobic glycolysis include biosynthetic materials such as pyruvate and lactate, which may provide the physical substrate for ongoing synaptic turnover.⁹⁴ Moreover, Goyal and colleagues⁹⁴ observed that the regional distribution of aerobic glycolysis in the cortex corresponds to regional transcription of juvenile genes ('neoteny')—especially those pertaining to synapse formation. This may at least partially explain why transmodal association cortices exhibit the highest synaptic density, as indicated by post-mortem analyses of cortical tissue⁹⁵ as well as *in vivo* imaging.^{96,97} Thus, at the top of the cortical hierarchy we find that (i) synapse formation is least inhibited by myelination; (ii) there is transcription of genes supporting synapse turnover; (iii) aerobic glycolysis makes continuously available the kind of biosynthetic materials that would support synaptic turnover; and (iv) there is the highest synaptic density.

The extended ability for transmodal association cortices to undergo neuroplastic change is also related to their slow rate of maturation. Whereas primary cortices reach adult-like spatial organization soon after birth, the apex of the cortical hierarchy continues to develop throughout childhood and adolescence⁹⁸ with heteromodal transmodal regions within the DMN being the last to reach full developmental maturation.⁹⁹ The maturation of transmodal association cortices in the human brain is also slow compared with corresponding cortical regions in non-human primates: e.g. in macaques and chimpanzees, the prefrontal cortex reaches its peak synaptic density in the first 12 months of life, but in humans this is only achieved around 5 years of age¹⁰⁰ and the greatest cortical surface area is found around the first decade after birth.¹⁰¹ Crucially, this prolonged period of maturation makes it possible for such regions to continue to be responsive to environmental influence and support ongoing learning through experience-dependent plasticity¹⁰²—a characteristic that is enabled by their high transcriptional and metabolic support for plasticity. In fact, evidence indicates that aerobic glycolysis increases during childhood in a manner that coincides with periods of highest synaptic growth.⁹⁴ Intriguingly, transmodal association cortices also exhibit the greatest degree of interindividual variability in functional connectivity patterns and spatial topography,^{3,6,103} reflecting the special sensitivity of these regions to diverse environmental influences.

The capacity for ongoing learning of the human brain is especially relevant because it has been shown that when organisms can learn during their lifetime, evolutionary paths can become available that would be foreclosed to non-learning organisms.¹⁰⁴ One especially powerful way that humans can benefit from the capacity for ongoing learning, afforded by the prolonged development and plasticity of transmodal cortex, is learning from conspecifics. As a highly social species with the unique ability to exchange information through language, humans can benefit from cumulative intergenerational learning (e.g. the invention of fire-making¹⁰⁵). Indeed, the 'Social Brain' account of human cognitive evolution highlights the need to adapt to the complex social dynamics arising from living in a group.¹⁰⁵ It is notable therefore that, in addition to

being highly flexible and plastic, the transmodal association cortices include the core brain regions that support social cognition.¹⁰⁶ This suggests that the transmodal cortex may be well poised to support cultural aspects of learning and evolution.

Developmental and evolutionary expansion of human transmodal cortex

Robust multimodal research has therefore revealed that a cortical hierarchy from unimodal to transmodal cortex constitutes the primary organizational axis of the cortex, based on a convergence of anatomical and functional evidence at both the micro- and the macroscale. Next, we review evidence indicating that this hierarchical organization also coincides with the pattern of cortical expansion across both ontogeny and phylogeny, with transmodal cortices exhibiting disproportionate expansion.

On the developmental side, although the brain as a whole expands substantially during humans' exceptionally protracted developmental period, transmodal association cortices expand by an approximate factor of four—twice as much as the expansion of primary cortices.³ This means that over the course of human brain development from birth to adulthood, transmodal association cortices come to constitute an increasing proportion of the total cortical volume—in correspondence with an increase in those cognitive capacities that are most distinctly human, such as executive control, abstract perceptually-decoupled thought and long-term planning. In other words, the progressive development of distinctly human cognitive capacities in human children coincides with the protracted ontogenetic expansion of the apex of the cortical hierarchy.

On the evolutionary side, similar conclusions about the role of transmodal association cortex in supporting distinctly human cognitive capacities can also be reached by comparing humans with non-human primates, such as the well studied macaque (*Macaca mulatta*, *M. fascicularis*) and the species most evolutionarily close to *Homo sapiens*: the chimpanzee (*Pan troglodytes*). Although the substantial differences between species and their unique environmental adaptations should not be underestimated when comparing cognitive abilities, it is evident that the range and complexity of cognitive aptitudes in humans far exceeds that of other mammals, including other primates. It is therefore reasonable to wonder what aspect(s) of the human brain most differentiate it from the brains of other primates. Even after accounting for differences in total brain size, humans exhibit disproportionate expansion of transmodal association cortices compared with other primates.^{4,6,107} Transmodal association cortices also notably express the highest rate of human-accelerated genes pertaining to brain function and development.⁴ Intriguingly, the regional prevalence of synergy (the super-additive gain in information that is present when two elements are considered together, such that the whole is greater than the sum of its parts^{108,109}) over redundancy (the extent to which regions are interchangeable in terms of the information they encode) also reaches its peak in the transmodal association cortex, correlating with a region's degree of evolutionary expansion and expression of human-accelerated genes.¹¹⁰ The overall reliance on synergy (but not redundancy) is also significantly higher in the brains of humans versus macaques¹¹⁰ providing additional evidence for its intimate link with higher-order cognition.

Taken together, multimodal research has revealed that a cortical hierarchy from unimodal to transmodal cortex constitutes the primary organizational axis of the cortex. In the next section,

we review the neurobiological underpinnings that underlie the exceptional expansion of transmodal cortex.

5-HT_{2A} receptors as potent modulators and key developmental drivers of human transmodal cortex

In the preceding sections we have established the unique structural and functional properties of transmodal cortex, highlighting its flexibility, plasticity and location at the apex of the cortical hierarchy. In what follows we review emerging research on 5-HT_{2A}R supporting its potent ability to modulate the functioning of human adult transmodal cortex and its potentially critical role in driving its developmental expansion.

Neuroanatomical localization of the 5-HT_{2A} receptor along the cortical hierarchy

The most reliable characterization of 5-HT_{2A}R spatial distributions in the adult human brain comes from high-resolution PET imaging studies, which used the 5-HT_{2A/2C} agonist radioligand ¹¹C-Cimbi-36.^{18,111} 5-HT_{2A}R distributions revealed by this radioligand correlate strongly with the more selective 5-HT_{2A} (antagonist) radioligand ¹⁸F-altanserin ($R^2 = 0.87$), while exhibiting greater test-retest reliability and sensitivity to high affinity receptor states.¹¹¹ *In vivo* human PET mapping with ¹¹C-Cimbi-36 has found that 5-HT_{2A}Rs are the most cortically expressed of all 5-HT receptor subtypes^{18,112,113} and, critically, that 5-HT_{2A}R densities are highest in transmodal cortex,^{18,111} with the overall receptor distribution recapitulating the unimodal-transmodal cortical hierarchy.¹⁸ We have quantitatively confirmed this visually-apparent spatial convergence (Fig. 1).

In addition to their high localization in human transmodal cortex, it is noteworthy that 5-HT_{2A}Rs, although expressed by both neurons and glial cells across layers, are especially enriched in layer 5 pyramidal neurons (L5Ps).^{115–119} L5Ps are the primary excitatory neurons of the cortex and are critical for information integration at both local and whole-brain levels. At the local level, their dendrites span all cortical layers, enabling them to integrate layer-specific feedback and feed-forward signals.^{120,121} At the whole-brain level, L5Ps exhibit long-range projections, which facilitate the integration of spatially distributed cortical and subcortical regions.^{120,121} As such, L5Ps—particularly those that reside in transmodal cortex—are well positioned to enable hierarchical information integration at both columnar and global scales, and thereby regulate global brain connectivity and dynamics.^{120,121} The localization of 5-HT_{2A}Rs on L5Ps within transmodal cortex therefore suggests that these receptors are poised to have a strong ability to modulate transmodal function and cortical hierarchical organization.

Basal progenitor cells, the 5-HT_{2A} receptor and uniquely human cortical expansion

Intriguing additional support for linkages between the 5-HT_{2A}R and transmodal cortex comes from recent research supporting a critical role for 5-HT and the 5-HT_{2A}R in particular, in the developmental expansion of human transmodal cortex. A large body of previous work has highlighted 5-HT as a critical regulator of neurodevelopmental processes.^{37,38} Pharmacological and transgenic studies to date have linked 5-HT to a variety of developmental processes, including neuronal differentiation, migration and

myelination, axonal guidance and synaptogenesis, and dendritic pruning.^{37,38,122–126} Several of these functions occur prior to the formation of synaptic circuits and therefore can be said to constitute ‘non-neurotransmitter’ roles for 5-HT. Indeed, evidence from the developing mouse brain indicates the presence of placental sources of 5-HT prior to the embryo's endogenous 5-HT delivered to the developing neocortex, during a time period that overlaps with multiple neurodevelopmentally critical events.¹²⁷ In addition, dysregulation of 5-HT signalling during early development (e.g. altered maternal 5-HT levels) has been strongly linked to the emergence of developmental and mood disorders, including autism, Down syndrome, generalized anxiety disorder and depression.³⁸ Critical to the present context, a recent multi-species study notably revealed that 5-HT, via 5-HT_{2A}R signalling, may be critical for evolutionarily-relevant processes, which underpin human cortical expansion and, by extension, the disproportionately expanded human transmodal association cortex.³⁵ We will now briefly provide important background prior to fully explicating this study.

Investigations of the mechanisms underlying cortical expansion have highlighted the importance of inter-species differences in cortical neurogenesis, a core neurodevelopmental process that hinges on the relative abundance and proliferative capacity of neural progenitor cells (NPCs).^{128–131} Comparative studies and studies using transgenic models have found that genetic alterations to distinct NPCs can, depending on the type of NPC targeted, result in distinct differences in cortical surface area, thickness and/or folding.^{34,128} Among the NPC types, so-called basal progenitor cells—and basal radial glia (bRG) in particular—exhibit marked differences across species and are particularly proliferative in gyrencephalic species, reaching their pinnacle in humans.^{128,132} bRG represent only 10% of neural progenitors in rodent species, whereas they represent ~50% in macaques and upwards of 75% in humans.^{131,133,134} Exceptionally high bRG abundance and proliferation has been specifically highlighted as a primary factor in uniquely human cortical expansion^{128,131} (Fig. 3).

Given the centrality of bRG in human cortical expansion, it is striking to note that 5-HT_{2A}R signalling during early development was found to be necessary and sufficient for the evolutionarily relevant proliferation of bRG in human, ferret and mouse cortical tissue.³⁵ Necessity was established by the finding that disruption of 5-HT_{2A}R in the embryonic ferret cortex specifically reduced the abundance of proliferative bRG.³⁵ Sufficiency was established by the finding that ectopic 5-HT_{2A}R expression in the developing lissencephalic mouse neocortex resulted in a 2-fold increase in the abundance of bRG.³⁵ Rounding the findings, application of a 5-HT_{2A}R agonist with high binding affinity (1 μ M NBOH-2C-CN)^{135,136} to human fetal cortical tissue *ex vivo* also resulted in a significant increase in proliferative bRG—an effect that was blocked by the administration of a 5-HT₂ receptor antagonist (EMD 281014).³⁵ The role of 5-HT_{2A}R is further supported by findings indicating a lack of 5-HT_{2A}R expression in neural progenitor cells of the developing lissencephalic mouse cortex, whereas 5-HT_{2A}R expression is evident in the developing gyrencephalic ferret and human cortex.^{35,124}

A relevant question is whether our hypothesis is specific to 5-HT_{2A}R, or whether it also applies to other serotonin receptors. With respect to 5-HT_{2A}R specificity in cortical expansion, the study by Xing et al.³⁵ used an antagonist (EMD 281014) with affinity for 5-HT_{2A}, B and C receptors, thereby excluding a necessary role for receptors beyond these. Importantly, both this antagonist as well as the agonist (NBOH-2C-CN) used in this study exhibit significantly higher affinity for human 5-HT_{2A} over 5-HT_{2C} and 5-HT_{2B}

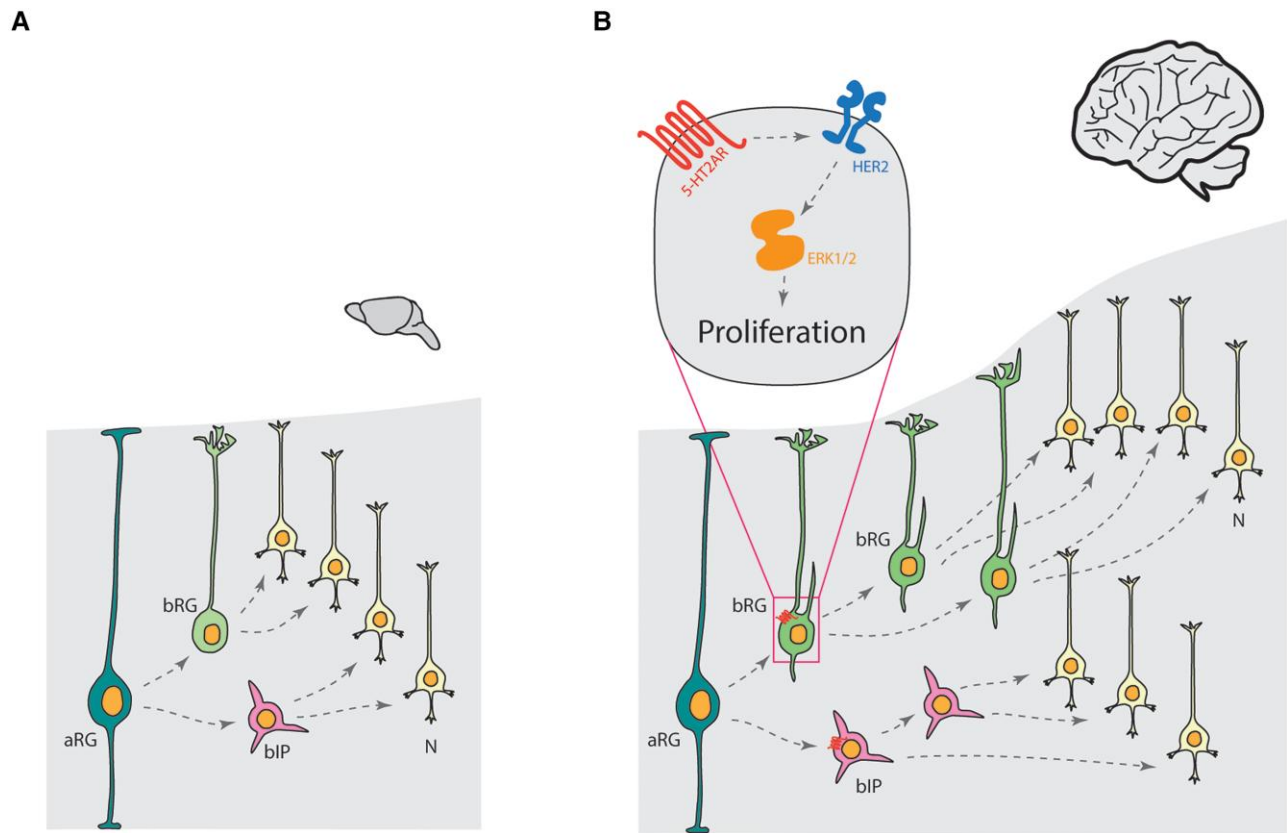


Figure 3 Model of how serotonin 2A receptor activation may contribute to the evolutionary expansion of the human neocortex. **(A)** Lineage relationships of neural progenitor cells in the developing mouse neocortex, where serotonin 2A receptor is absent. **(B)** Lineage relationships of neural progenitor cells in the developing human neocortex, where serotonin 2A receptor activation promotes the proliferation of basal progenitors such as basal radial glia (bRG) and basal intermediate progenitors (bIPs) via HER2 and ERK1/2 signalling pathways.³⁵ The increases in the abundance and proliferative capacity of basal progenitors lead to increased neuron (N) production and the expansion of the human neocortex.¹²⁸ aRG = apical radial glia.

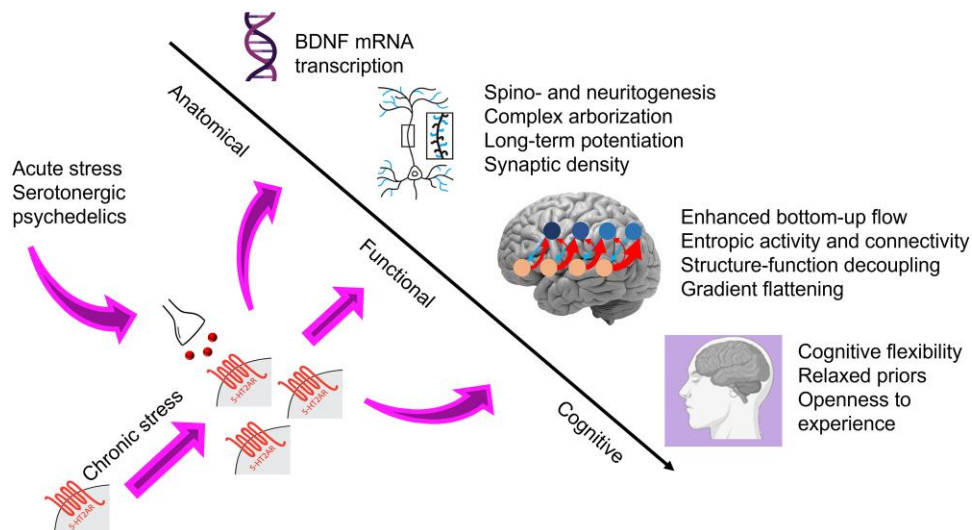


Figure 4 5-HT_{2A}R-mediated anatomical, functional and cognitive plasticity. A schematic displaying two sources of 5-HT_{2A}R agonism (endogenous 5-HT release via acute and chronic stress and agonism by serotonergic psychedelics), as well as the putative primary anatomical, functional and cognitive effects of such agonism. Chronic stress primes the brain by increasing expression of 5-HT_{2A}Rs and their sensitivity to signalling. The primed 5-HT_{2A}R system can then be engaged by acute stress (which potentially releases 5-HT) or by serotonergic psychedelics. Effects on plasticity can then be observed across scales, from the molecular to the cognitive level. BDNF = brain-derived neurotrophic factor. Figure parts adapted from Luppi et al.³²⁸ and Vargas et al.³⁰⁹ (both under CC-BY license).

receptors (~ 8× for EMD 281014 and ~100× for NBOH-2C-CN¹³⁷). In addition, transcripts for 5-HT1A, 1D, 1E, 2B, 2C and 4 were not found in human basal NPCs, corroborating the specificity of these results (and our hypothesis) for 5-HT2AR.³⁵

Although direct empirical evidence is needed to establish a causal role, the high density of 5-HT2ARs in adult transmodal cortex suggests that the additional neurons generated in humans as a result of 5-HT2AR-mediated increases in proliferative bRG may be those that go on to comprise the transmodal cortices. This is consistent with the timing of distinct neurogenic phases during ontogenetic cortical development.¹³⁸ In particular, studies have indicated that neurogenesis via bRG comprises late-stage neurogenesis and that, in addition to laterally expanding the cortical surface, it results in a radial expansion of cortex via an increase in upper layer neurons.^{128,131} One would therefore expect that the regions of cortex that have the greatest basis in bRG-related neurogenesis would (i) be the last to develop; and (ii) have greater cortical thickness relative to other areas. Both of these properties have been observed to be the case in human transmodal cortex.^{1,52,53} Collectively, these findings suggest that 5-HT signalling at 5-HT2ARs in the embryonic/fetal brain may contribute to create the expanded transmodal cortices that, in adulthood, densely express 5-HT2AR receptors.

Role of other neurotransmitter systems and receptors in cortical expansion

In addition to 5-HT, basal progenitor abundance and proliferation—and, by extension, cortical expansion—is regulated by a variety of cell-extrinsic molecular factors.^{34,128} Among the most well characterized of these factors are extracellular matrix components, growth factors, thyroid hormones and neurotransmitters—each of which have been linked, via varying mechanisms, to increased NPC abundance and proliferation in developing cortex.^{139–143} Amongst the neurotransmitters, glutamate and GABA have been most studied for their effects on NPC proliferation.^{36,144–146} Both of these neurotransmitters regulate NPC proliferation through several distinct mechanisms, in a manner that appears to depend on the species, cortical region in question, and environmental context.³⁶ For example, activation of the AMPA/kainate glutamate receptor decreases NPC proliferation in germinal zones in developing rat cortical tissue,¹⁴⁵ whereas glutamate NMDA receptor agonism decreases NPC proliferation in the mouse cortex but increases proliferation in fetal human cortex.¹⁴⁷ With respect to GABA, GABA-A receptor agonism has been found to reduce the proliferation of apical progenitors in the ventricular zone in rat cortex,¹⁴⁵ whereas GABA-A and GABA-B agonism has been found to increase the proliferation of certain NPCs in mouse cortex.^{148,149} Important for the present context, although there is heterogeneity in the manner in which glutamate and GABA affect NPC proliferation across (and within) species, their ability to regulate NPC proliferation in general is conserved across both lissencephalic and gyrencephalic species.³⁶ In contrast, 5-HT2AR are absent on mouse NPCs but highly expressed on the NPCs of humans¹²⁴ and, as described, selectively stimulate the proliferation of bRG that are instrumental for human cortical expansion.^{35,128} As such, research to date on neurotransmitter contributions to cortical expansion suggest that, with respect to other neurotransmitters, 5-HT, via 5-HT2A agonism, may play an especially prominent role in the disproportionate expansion of transmodal cortex in humans.

5-HT2AR agonism in the adult brain: structural, functional and behavioural effects

Having reviewed neuroanatomical evidence in support of 5-HT2AR's potent ability to modulate transmodal functioning in the adult brain, as well as its potential critical role in the developmental expansion of transmodal cortex, we now discuss research on the structural, functional and behavioural effects of 5-HT2A agonism. We begin with a brief overview of the neuronal effects of 5-HTAR receptor agonism, followed by a discussion of conditions that favour endogenous 5-HT2AR agonism, and then review studies of pharmacologically-induced 5-HT2AR agonism via 5-HT2A agonist psychedelic drugs. We examine both the acute and longer term effects of 5-HT2AR agonists on brain structure, function and behaviour, highlighting a recurrent common theme: increased plasticity and flexibility, where plasticity is defined as the ability of a phenomenon (e.g. brain or behaviour) to be shaped or moulded—or, more plainly, to change (Fig. 4).

Neuronal effects of 5-HT2AR agonism in the adult human brain

The 5-HT2A receptor is an excitatory G-protein coupled receptor, with 5-HT2A agonism activating distinct intracellular cascades via G_q and arrestin signalling pathways.^{150–152} Electrophysiological studies have found that 5-HT2AR agonism has the net effect of increasing neuronal excitability as a result of downstream effects on glutamatergic neurotransmission.^{153,154} In particular, endogenous 5-HT2AR activation by serotonin has been found to increase both the amplitude and frequency of excitatory postsynaptic potentials in cortical layer 5 pyramidal cells.^{154,155} This was found to be via an 'asynchronous' mode of glutamate release that results in a relatively sustained enhancement of excitatory currents.¹⁵⁴ Interestingly, electrophysiological evidence suggests that pharmacological 5-HT2AR agonism via serotonergic psychedelic drugs leads to a unique set of neuronal effects via a combination of differential G protein/arrestin recruitment and access to intracellular 5-HT2ARs.^{156,157} These effects notably include the induction of recurrent loops of activation within a subset of deep layer 5 cortical pyramidal cells.^{153,155,158} The resulting recurrent loops appear to result in a diffuse mode of glutamate release which, via volume transmission effects, contribute to the local dysregulation of neuronal populations.^{153,159} Consistent with this, magnetoencephalography (MEG) studies with psilocybin and LSD, as well as an EEG study with *N,N*-dimethyltryptamine (DMT), have revealed broadband reductions in oscillatory power across most of the cortex, with peak reductions notably found in transmodal regions, such as the posterior cingulate.^{160–162}

In general, the neuronal effects of 5-HT2AR agonism, combined with their high density on layer 5 pyramidal cells within transmodal cortex, suggest a particularly potent ability to modulate transmodal function and global brain dynamics. Evidence suggests that this may particularly be the case for 5-HT2AR agonist psychedelic drugs—a notion also supported by a rapidly growing body of functional MRI evidence (reviewed later) indicating that acute 5-HT2AR agonism via such drugs induces significant alterations to global brain connectivity and dynamics, centred largely on changes to transmodal cortex.^{160,161,163–166}

5-HT_{2A}R agonism via endogenous 5-HT: the central role of stress

Serotonergic innervation in the adult human brain is predominantly provided by afferents originating in the raphe nuclei of the brainstem and evidence to date suggests that these nuclei—spanning dorsal, medial and magnus subdivisions—collectively release serotonin across nearly every cortical region, with relatively low regional specificity of innervation.^{167,168} (Although, it should be noted that there is spatial selectivity in the projections of distinct groups of raphe neurons.¹⁶⁹) As such, complexity in serotonergic modulation of cortical function is understood as predominantly emerging from the distinct characteristics (e.g. ionotropic versus metabotropic, differential G protein activation, high versus low affinity) and spatially heterogeneous distributions of serotonin receptor subtypes, rather than regional variation in levels of serotonergic innervation *per se*.^{169–172}

Of the serotonin receptor subtypes, 5-HT_{1A} and 5-HT_{2A} are the most abundantly expressed in the brain.¹⁸ Notably, serotonin has significantly higher affinity for 5-HT_{1A}R relative to 5-HT_{2A}R¹⁷³—suggesting that significant 5-HT_{2A}R agonism may only occur in the context of exceptionally high 5-HT. Of the variety of behavioural and physiological factors that regulate 5-HT release,^{169,170,172} perhaps the most powerful and reliable means of increasing neural 5-HT levels and engaging the 5-HT_{2A}R system is via stress—an organism's multi-system (allostatic) response to homeostatic challenge.^{174–176} Mild stress may have healthy 'hormetic' effects, e.g. stretching an organism's physiological range and associated resilience (e.g. as with intermittent moderate exercise)¹⁷⁷ but intense, repeated stress may be the cause of a major state transition, as in 'allostatic overload'¹⁷⁸ and so-called 'pivotal mental states', i.e. transient hyperplastic states conducive to psychological transformation.¹⁷⁵

Evidence indicates that the effects of stress on the 5-HT_{2A}R system are 2-fold. First, chronic stress increases cortical 5-HT_{2A}R expression and sensitivity to signalling^{179–182} (see Brouwer and Carhart-Harris¹⁷⁵ for a recent review). Such effects can be observed in response to physiological stressors such as deprivation of oxygen,¹⁸³ deprivation of sleep^{184–186} and inadequate nutrition,¹⁸⁷ and also in response to social/cognitive stress, such as recurring defeat,¹⁸⁸ rearing in isolation^{189–191} and maternal separation.^{192,193} Physiological or social deprivation may be a common factor here, with a 'priming' of the 5-HT_{2A}R system occurring as an allostatic response to these environmental deficiencies.

Second, it is well established from both human and rodent studies that acute stress reliably acts as a potent trigger for 5-HT release, e.g. in response to tail pinch, handling and swim stress,¹⁹⁴ fasting,^{195–197} acute social defeat^{198–202} and acute pain,²⁰³ with regional PET ¹⁸F-altanserin binding co-varying with pain responses in humans.²⁰⁴ Acute stress has also been found to promote the plasticity marker brain-derived neurotrophic factor (BDNF) in the prefrontal cortex.²⁰⁵ Other work has shown that BDNF is robustly and selectively increased in the cortex after 5-HT_{2A}R agonism.²⁰⁶

In past work, we synthesized a large body of findings (partially reviewed later) and argued that 5-HT_{1A}R agonism during times of low/intermediate 5-HT may facilitate a passive coping style—wherein individuals become more patient and less anxious in the face of adverse circumstances—whereas 5-HT_{2A}R agonism during times of high 5-HT facilitates an active coping style, which involves adaptively and flexibly responding to the challenges at hand.^{174,175} Thus, we proposed that low-grade/chronically stressful situations that might be adequately dealt with passively are underpinned by

lower serotonin levels and a relative dominance of 5-HT_{1A}R agonism, whereas intense, acutely stressful experiences (or chronic stress paired with an acute event) may demand a more active and adaptive behavioural response that is underpinned by higher serotonin and 5-HT_{2A}R agonism.^{174,175}

This notion of distinct behavioural strategies based on relative neural concentrations of serotonin is also consistent with a framework recently proposed by Shine *et al.*¹⁷¹ These authors argued that states of low/intermediate cortical serotonin are dominated by non-5-HT_{2A}R serotonergic innervation of the cortex via the cerebellum, and that this leads to behaviour that is driven by computationally cheap cerebellar automatisms.¹⁷¹ In contrast, according to this proposal, during circumstances in which automatized cerebellum-based behaviours do not suffice to address environmental challenges (such as, we argue, during times of significant stress), central serotonin concentrations will be increased to a level sufficient to engage 5-HT_{2A}R and thereby shift the balance towards cortical computation with greater flexibility and adaptability.¹⁷¹

On the whole, a picture emerges whereby chronic stress may be seen as 'priming' the 5-HT_{2A}R system as part of the organism's response to a situation of deprivation/stress (physiological, social or even sensory), and then this primed system can then be activated by potent 5-HT release upon acute stress.¹⁷⁵ The consequent high levels of 5-HT high levels of then lead to the engagement of the 5-HT_{2A}R system, which in turn facilitates an adaptive and flexible behavioural and cognitive style aimed at actively responding to the demands of the current environment. Critically, this conception of 5-HT and 5-HT_{2A}R largely accords with research on exogenous agonism of 5-HT_{2A}R via psychedelic drugs, which has found evidence of increased structural, functional and behavioural flexibility. We review this research next.

Exogenous 5-HT_{2A}R agonism via psychedelic drugs

Further insight into the effects of 5-HT_{2A}R agonism in the adult human brain comes from work with 5-HT_{2A}R agonist psychedelic drugs.^{17,207} Such compounds include the naturally occurring substances DMT, 5-methoxy-DMT (5-MeO-DMT), psilocybin and its metabolite, psilocin, the psychoactive component of psilocybe 'magic mushrooms', as well as the peyote-derived, mescaline. Synthetic psychedelics include LSD, but also the phenethylamines 2,5-dimethoxy-4-iodoamphetamine (DOI) and 2,5-dimethoxy-4-bromoamphetamine (DOB). 3,4-Methylenedioxy-methamphetamine (MDMA) is sometimes regarded as a psychedelic, given its profile of subjective effects and its agonist effects at 5-HT_{2A}R. However, rather than displaying direct agonist properties at 5-HT_{2A}R, it only indirectly stimulates these receptors via potent 5-HT release.¹⁹⁶ Psychedelic drugs have been found to induce a complex variety of subjective effects, spanning changes to perception, cognition, emotion and sense of self.^{17,208,209} A growing body of evidence also suggests their efficacy—when combined with supportive psychotherapy—in the treatment of several mental health conditions.²¹⁰

In both humans and other animals, the potency of a given psychedelic's effects on subjective experience, cognition and behaviour can be reliably predicted from its affinity for 5-HT_{2A}R at the receptor.²¹¹ In rodent models, the subjective effects of psychedelics (as indicated by head twitching) are specific to agonism of 5-HT_{2A}R and can be selectively blocked by 5-HT_{2A}R antagonists.²¹² In humans, psychedelics elicit a wide range of subjective effects and significant alterations to global brain function, both of which are blocked by pretreatment with the 5-HT_{2A}R antagonist, ketanserin.^{213–215} Consistent with this, a recent PET study notably

found that the intensity of subjective effects induced by psilocin, the active metabolite of psilocybin, significantly correlated with 5-HT_{2A}R occupancy.²¹⁶

Taken together, there is convergent neural and behavioural evidence, from humans and animal models, as well as computational studies, that 5-HT_{2A}R agonism is both sufficient and necessary to account for the complex effects of psychedelic agents and constitutes their main pathway of action. Psychedelic-specific 5-HT_{2A}R signalling cascades appear to exist^{157,112} and the expression of plasticity genes has been implicated.²¹⁷

Anatomical neuroplasticity induced by 5-HT_{2A}R agonism

The acute effects of 5-HT_{2A}R agonism are evident at the neuroanatomical level. Evidence from mice deprived of vision from one eye indicates that 5-HT_{2A}Rs are required for cross-modal recruitment of monocular cortical territory by the whiskers, a form of plasticity that can occur even in the adult brain—and which is abolished by pharmacological antagonism of 5-HT_{2A}Rs, but not 5-HT_{1A}Rs.²¹⁸ Additionally, early work in rodents demonstrated that treatment with the partially selective 5-HT_{2A}R agonist, DOI, produced a doubling of mRNA expression pertaining to BDNF within the cortex.²⁰⁶ Although DOI is not selective between 5-HT₂ receptor subtypes, this effect was shown to be specifically mediated by 5-HT_{2A}R agonism, since it could be prevented by pretreatment with the 5-HT_{2A}R antagonist ketanserin, but not with a 5-HT_{2C}R antagonist (Box 1).

Even more direct evidence of psychedelic-induced increases in neuroanatomical plasticity was provided by Jones et al.,²²⁶ who reported that 5-HT_{2A}R agonism by DOI can induce a transient increase in the size of dendritic spines of rat cortical pyramidal neurons. More recently, compelling work by Ly and colleagues²⁹ showed that LSD, DMT and DOI significantly increase the complexity of dendritic arbours and promote neuritogenesis and spinogenesis. Corroborated by 5-HT₂ receptor antagonist tests with ketanserin, overall the results of this study suggest 5-HT_{2A}R receptor involvement in psychedelic-induced neuroanatomical plasticity, further supported by the observation that the growth of dendritic spines and synapses induced by a given compound correlated with its affinity for the serotonin 2A receptor²⁹ (Box 1).

Complementing these findings, longitudinal two-photon imaging of layer 5 pyramidal neurons within mouse frontal cortex (a major locus of 5-HT_{2A}R expression) revealed that psilocybin can induce significant increases in the size and density of apical dendritic spines.³¹ This effect, which could be induced within 24 h of administration of a single dose of the psychedelic, was found to persist after 1 month (although this effect was only observed in female animals, warranting further investigation), and coincided with a decrease in the rodents' behavioural manifestation of stress responses.³¹ This supports a link between 5-HT_{2A}R agonism and long-lasting and behaviourally relevant neuroplastic change. In addition, a recent study applied a novel *in vivo* measure of synaptic density from PET imaging and demonstrated that psilocybin can increase synaptic density in the brain of pigs, concomitantly with the well documented reduction in 2A receptor density that follows its acute engagement by psychedelics.³⁰

Functional neuroplasticity induced by 5-HT_{2A}R agonism

In addition to the abovementioned anatomical changes, functional changes can also be identified as a result of acute 5-HT_{2A}R

agonism. Functional MRI studies with LSD, psilocybin and DMT have found that they induce a mode of brain function that features greater integration between and reduced integration within the majority of large-scale brain networks.^{161,163,165,166,215,227–229} These changes are in alignment with the regional distribution of 5-HT_{2A}Rs as revealed by *in vivo* PET imaging^{166,228}, although see Preller et al.^{215,227} Going beyond correlation, *in silico* studies using network control theory²³⁰ or dynamic mean-field models of coupled excitatory and inhibitory populations,^{79,164,220,231,232} have shown that the effects of LSD and psilocybin on both local and global brain dynamics can be modelled mechanistically by including the regional distribution of 5-HT_{2A}Rs (but not other serotonin receptors).

Leading theoretical accounts of psychedelic action that aim to reconcile neural findings with the subjective and therapeutic effects of these drugs, such as the 'Entropic Brain Hypothesis'^{233,234} and its recent evolution, the 'Relaxed Beliefs Under Psychedelics' (REBUS) model,³² postulate that the principal acute functional action of 5-HT_{2A}R agonist psychedelics is the dysregulation of spontaneous, population-level cortical activity, manifesting as an increased complexity or entropy of spontaneous brain activity.^{32,162,233–235} In virtue of 5-HT_{2A}R localization at the apex of the cortical hierarchy in transmodal cortex, this dysregulation is thought to predominantly result in disruption of top-down predictive processing.^{32,233,234,236,237}

The serotonergic psychedelics LSD, psilocybin and DMT have also been found to increase the diversity (quantified as entropy or incompressibility) of regional brain activity and functional connectivity over time whether measured with electro- or magneto-encephalography^{160,235,238} or functional MRI,^{221,239–247} including one study that found the entropic effect to be predictive of subsequent psychological changes.²³⁹

Complementing these various lines of evidence, a recent study based on the theory of optimal control recently revealed that both LSD and psilocybin induce a 'flattening' of the brain's energy landscape, corresponding to reduced energy required to transition between distinct patterns of whole-brain activity, making such transitions more fluid.¹⁶⁴ Across the various serotonin receptors, 5-HT_{2A}Rs are uniquely well suited to induce a reduction of the brain's optimal control energy—indicating that 5-HT_{2A}R agonism is likely to be the key triggering mechanism accounting for the empirically observed effect.²⁴⁸

LSD and psilocybin have also been found to induce a decoupling between neural structure and function. More specifically, recent studies have observed that these two drugs induced a dissociation between the brain's macroscale network of white matter structural connections (the human connectome) and the patterns of functional activity^{243,244} and connectivity²⁴⁶ that unfold over it. In light of Hebb's well known dictum that 'neurons that fire together, wire together', the brain's macroscale structural connectivity may be viewed as encoding evolutionary and developmental expectations (or 'priors') about which regions should preferentially communicate with each other. In turn, the psychedelic-induced decoupling of function from structure can then be interpreted as reflecting a deviation from such predetermined patterns in favour of broader exploration, i.e. analogous to a 'journey' or 'trip' away from well-trodden paths.

A diminished influence of top-down information processing has been reported across several psychedelics and diverse investigative strategies. Specifically, an investigation of cortical traveling waves showed that DMT attenuates the top-down alpha-band EEG rhythms that usually characterize the resting brain, in favour of

Box 1 Specificity of psychedelic effects for the 5-HT_{2A} receptor

Pertaining to both the neural and subjective effects of psychedelics, their abolition via ketanserin pretreatment has excluded a primary causal role of receptors beyond the 5-HT₂ group.^{207,213,215} In mice, the head-twitch response to psychedelics can be abolished via genetic knockout of 5-HT_{2A}Rs.^{112,219} In humans, the preferential involvement of the 2A receptor is further (albeit indirectly) corroborated by computational studies showing that 2A expression maps provide better fit to the neural effects of LSD and psilocybin than 5-HT_{1A}, 5-HT_{1B} and 5-HT₄ maps, as well as dopamine D1 and D2 receptor expression.^{220,221} However, ketanserin is a non-selective antagonist of 5-HT₂ receptors: although it has 30-fold selectivity for 5-HT_{2A}R over 5-HT_{2C}R,²²² these results cannot rule out 5-HT_{2C}R involvement.

Pertaining to 5-HT_{2A}R involvement in promoting neuroanatomical plasticity, both the study by Vaidya and colleagues²⁰⁶ and the recent investigations by Jones and colleagues²²⁶ and Ly and colleagues²⁹ showed that increased markers of plasticity (BDNF mRNA, dendritic spine size, and neuritogenesis and spinogenesis) could be observed after treatment with DOI, which is a highly selective agonist for 5-HT₂ receptors over all other G-protein coupled receptors. Vaidya *et al.* and Ly *et al.* additionally showed that DOI-induced increases in neuroplasticity were abolished by ketanserin, and Vaidya and colleagues further excluded a role of 5-HT_{1A}R, since its agonist 8-OH-DPAT produced no effect. On their own, these results strongly implicate 5-HT₂ receptor agonism as both necessary and sufficient for inducing markers of plasticity in rodents. Adding to this, the seminal study by Vaidya and colleagues²⁰⁶ was able to demonstrate 5-HT_{2A}R specificity over 5-HT_{2C}R: they found that DOI regulation of BDNF mRNA expression is completely abolished by pretreatment with MDL 100907, which has a 100-fold greater affinity for 5-HT_{2A}R than 5-HT_{2C}R.¹⁶⁶ In contrast, the authors still observed DOI-induced increase in BDNF mRNA expression after pretreatment with SB 206553, which has a 100-fold preference for 5-HT_{2C}R over 5-HT_{2A}R.^{223,224} Thus, the results of this study converge on 5-HT_{2A}R agonism in the regulation of plasticity.

Finally, we note that multiple serotonergic Gs-linked receptors—representing a distinct family of G protein-coupled receptors than 5-HT_{2A}R—are present in the human brain; namely, the 5-HT₄, 5-HT₆ and 5-HT₇ receptors.²²⁵ Although these receptors are central to endogenous 5-HT signalling in the adult human brain, there is no evidence that these receptors are expressed in neural progenitor cells during cortical development¹²⁸ and we therefore do not focus on them in the present review.

Overall, there is evidence from a variety of investigative approaches strongly implicating 5-HT₂ receptor agonism in basal progenitor cell proliferation during development, as well as adult neural plasticity in rodents, and the subjective and neural effects of psychedelics in humans—over and above other neurotransmitters, and other types of serotonin receptors. Additionally, the results suggest a preference for the 2A over 2C receptor, although the evidence is less definitive in this regard.

waves traveling from the bottom up.^{229,236} Diverse methods to infer the directionality of connectivity between brain regions, including dynamic causal modelling and transfer entropy, have also consistently identified diminished top-down influences, using MEG and EEG^{160,236,249,250} and functional MRI.²⁵¹

Highly relevant to the present paper's main focus, fMRI research has also shown that LSD and psilocybin induce a 'flattening' of the cortical unimodal-transmodal functional hierarchy—as indexed via the principal sensorimotor-to-association functional⁸—by increasing cross-talk between these usually relatively segregated functional zones.^{161,163,166,228,229} As we have highlighted, this gradient corresponds to the spatial distribution of 5-HT_{2A}Rs across the cortex. Therefore, this study demonstrates that acute 5-HT_{2A}R agonism can modulate the brain's macroscale cortical functional hierarchy in the adult brain. Importantly, hierarchical functional organization is necessary for the instantiation of hierarchical predictive mechanisms, thought by many to be a key operative mechanism of the brain.^{252,253} Thus, by implication, if the brain's main hierarchical gradient is 'flattened' or 'compressed' under psychedelics, top-down predictive mechanisms should be compromised—consistent with the REBUS model. One possible interpretation of this effect is that bottom-up information flow, i.e. 'prediction error', will be liberated to impress on supraordinate regions and systems—potentially driving the updating of predictive encodings (i.e. the 'posterior distribution').³² Evidence for the revision of high-level models or beliefs post-psychedelic use can be seen here²⁵⁴—but there are multiple other ways in which this effect may express itself. Indeed, the revision of pathological predictive encodings is hypothesized to be a key component of the therapeutic action of psychedelic therapy.²⁵⁵

Cognitive and behavioural plasticity induced by 5-HT_{2A}R agonism

Mounting evidence indicates that flexibility of cognition and behaviour in the face of environmental changes are mediated by serotonin.^{256–260} Conversely, behavioural flexibility is impaired in marmoset monkeys following experimental depletion of serotonin from the orbitofrontal cortex, resulting in perseverative behaviour.²⁶¹ Evidence for a role of 5-HT_{2A}R in mediating the relationship between serotonin and cognitive flexibility comes from animal models: 5-HT_{2A}R agonists such as LSD can improve the ability of non-human animals to learn novel associations.^{262,263} In humans, evidence for increased learning capacity comes from a recent study combining acute pharmacological intervention with LSD and computational modelling of trial-and-error reinforcement learning, which found that subjects had an increased ability to update the expected value of performing a given action based on feedback.²⁶⁴ Both the subjective (psychedelic) and neural effects of LSD in humans can be blocked by pretreatment with ketanserin^{207,213,215}; however, given the agonism of LSD for dopamine (albeit substantially weaker) and for serotonin receptors beyond 5-HT_{2A}R, and the involvement of dopamine in reinforcement learning,^{265,266} future work will be needed to conclusively establish whether these computational effects are also uniquely attributable to 5-HT_{2A}R agonism.

This computational evidence is in line with additional evidence that acute administration of ayahuasca induces a shift in cognition away from convergent and towards divergent modes of thinking.²⁶⁷ Similar findings suggest that LSD modulates creativity towards novelty and a larger semantic spread.^{268–270} Such changes in

cognitive style need not be confined to the acute experience; indeed, the evidence for an acute action of 5-HT_{2A}R manipulation on divergent thinking and cognitive flexibility is somewhat mixed.^{271–274} Post-acute increases in markers of cognitive flexibility appear to be more reliable. For example, a recent study of patients suffering from major depressive disorder found that subacute increases in cognitive flexibility were present at 1-week post-session and maintained for at least 4 weeks.²⁷¹ In addition, a study of the effects of psilocybin on common creativity tasks found evidence for post-acute improvements, with an absence of improvements acutely.²⁷² These studies are further consistent with additional studies indicating post-acute ‘after glow’ effects of increased cognitive flexibility and creativity.^{275,276} Another, albeit indirect, source of evidence comes from investigations of personality change following psychedelic administration. Investigations with psilocybin have revealed significant increases in the personality domain of openness to experience at long term follow-ups in both healthy subjects²⁷⁷ and patients suffering from treatment-resistant depression.²⁷⁸ Increased openness was also reported 2 weeks after LSD administration in healthy subjects, an effect that could be predicted from functional MRI measures of entropy LSD administration²³⁹—providing preliminary evidence for a bridge between acute functional complexity and enduring cognitive plasticity. Finally, studies have also found psilocybin-induced post-acute increases in psychological flexibility, a therapeutically-relevant construct derived from acceptance and commitment therapy (ACT) that relates to one's ability to flexibly respond to the present moment.^{279,280}

A recent account of serotonin's multi-faceted role in neural computation proposed that serotonin concentration may track the availability of time and resources, and whether the present state is generally beneficial.¹⁷⁰ According to this account, greater availability of time (signalled by high serotonin concentration) would allow for perception to be based more on incoming sensory evidence and less on priors¹⁷⁰—consistent with our proposed account and a psychedelic-induced ‘weakening of priors’ mediated by 5-HT_{2A}R engagement.³² The same account also proposes that greater serotonin would promote slower learning rate, given that this would coincide with more time for learning and a consequent more exhaustive (wide and/or deep) exploration of what is being learned. At initial glance, this stands in apparent contrast with the evidence for the psychedelic and stress-induced enhancement of learning and plasticity reviewed above. However, we note that whereas our proposed account highlights an increased ability for the brain to structurally and functionally adapt to new environments and contexts, this account highlights a possible serotonin-facilitated behavioural inclination to engage in a wider search and collect a greater amount of evidence during learning. As such, these accounts are not mutually exclusive and raise interesting testable hypotheses pertaining to how serotonin/5-HT_{2A}R agonism might differentially alter ‘absolute’ learning rate (via plasticity promotion), the breadth of learning that is naturally pursued (as a result of relaxed priors and a perception of more available time) and the ratio between the two.

Therapeutic applications of 5-HT_{2A}R agonism

Abnormalities centred on transmodal association cortex have been implicated in a range of psychiatric conditions, as recently reviewed by Sydnor et al.¹ A large and growing literature has identified structural (e.g. reduced volume and thickness) and functional (e.g. changed large-scale network connectivity) alterations in transmodal cortices, which characterize individuals suffering from diverse psychiatric

symptoms, from anxiety and depression to generalized psychopathology and psychosis.^{281–283} In addition, genes pertaining to the organization of association cortex are implicated in genetic vulnerability to a host of psychiatric disorders.^{284,285} Given the centrality of transmodal cortex in psychopathology and given the above-reviewed neuroanatomical and functional characteristics of 5-HT_{2A}Rs, it is reasonable to hypothesize that 5-HT_{2A} agonist drugs may have therapeutic relevance. Notably, this is supported by recent clinical trials supporting the efficacy of psychotherapeutic interventions involving serotonergic psychedelic drugs for several mental health conditions²⁸⁶ (see [Supplementary Table 1](#) for a summary of trials to date). Evidence indicating beneficial effects of 5-HT_{2A}R agonism on clinical symptomatology and/or well-being has been steadily accumulating¹⁹ from investigator-initiated clinical trials^{20–22,24–27,287–296} ([Supplementary Table 1](#)) and controlled studies in healthy individuals.^{297–300} Prospective surveys of naturalistic use have also found increased subjective well-being after the psychedelic experience^{301–304} (reviewed in Johnson et al.¹⁹), even 2 years later.³⁰⁵ The quality of evidence supportive of psychedelic-assisted psychotherapy was recently bolstered by the publication of high-profile clinical trials of MDMA therapy for post-traumatic stress disorder,³⁰⁶ and psilocybin-therapy for major depressive disorder.^{20,22} Consistently high response rates exceeding 70% were seen across all three of these studies in those treated with psychedelic-assisted psychotherapy.

A core characteristic of psychedelic treatments for mental health is their dependence on extra-pharmacological factors and their administration in the context of adjunctive psychotherapeutic support (hence, ‘psychedelic-assisted psychotherapy’^{307,308}). The context-dependence of the therapeutic action of psychedelics dovetails with the evidence presented in previous sections supporting a close association between increased neuroplasticity and corresponding therapeutic effects.^{33,309,310} In particular, we and others have highlighted how 5-HT_{2A}R-induced plasticity is itself agnostic with respect to outcomes: whether or not neuroplastic changes are ‘therapeutic’ (i.e. supportive of positive mental health) is dependent on the nature of the contextual factors present prior to, during and following drug administration.^{175,311} One way this may be described is that psychedelic-assisted psychotherapy, via 5-HT_{2A}R agonism combined with therapeutic support, may temporarily increase and harness the capacity of transmodal cortex—highly active during development—to be moulded by socio-cultural/environmental learning, in order to facilitate adaptive and health-promoting neuroplastic changes.^{175,308} This idea also closely parallels recent work which found that psilocybin, MDMA and other psychedelics open critical periods for social reward learning, providing evidence that temporarily enhanced social learning (via temporarily increased plasticity) may contribute to the therapeutic effects of psychedelic-assisted therapy.^{312–314}

If 5-HT_{2A}R agonism exerts its beneficial effects by enhancing neural and psychological plasticity, then a possible synergy becomes apparent with psychotherapeutic techniques that emphasize a flexible, accepting approach to one's emotions, memories and circumstances, such as mindfulness-based therapies and ACT in particular.^{32,279,280,315–317} Such psychotherapies may marry well with psychedelics due to their ability to harness the enhanced plasticity triggered by the drugs' pharmacological action, as reviewed in the previous sections. Nevertheless, we emphasize that complex psychiatric conditions such as PTSD, major depressive disorder and addiction are invariably the result of intricate interactions between a patient's neurobiology, cognition and environment, and are therefore best addressed as such; hence our focus is on the potential for psychedelics to enhance and facilitate psychotherapeutic processes, rather than being pure pharmacotherapeutic agents.

Role of other neurotransmitter systems and receptors in plasticity

It is important to emphasize that, although the 5-HT_{2A}R system is involved in plasticity and flexibility at the anatomical, functional and cognitive levels, it is by no means the only plasticity-related system. It is well known that glutamatergic signalling involving AMPA and NMDA receptors plays a key role in long-term potentiation and long-term depression of synapses³¹⁸ both in terms of enacting short-term changes in synaptic strength and ensuring their long-term maintenance through regulation of gene expression.³¹⁹ Indeed, evidence suggests that 5-HT_{2A} agonism induces its pro-plasticity effects via its downstream effects on glutamatergic neurotransmission.^{159,320} Other neuromodulators than serotonin also shape plasticity, with both convergent and divergent roles.³²¹ Dopamine has been robustly associated with reward prediction errors, providing a mechanism to address the problem of ‘credit assignment’³²²: which connections should be changed and how, to reduce the difference between expected and observed reward?³²³ Whereas dopamine (and to some extent noradrenaline) regulates plasticity after-the-fact in response to ‘unexpected uncertainty’,³²¹ acetylcholine may facilitate plasticity proactively in the presence of ‘expected uncertainty’, by controlling vigilance and selective attention, which are widely known to enhance learning.³²³ In addition, GABAergic inhibition has been shown to control the critical window of plasticity during development,³²⁴ and blocking GABAergic signalling can restore the plasticity of sensory cortex in adult animals.³²⁵ Since the duration of the critical period is greater in humans than other primates,¹⁰⁰ especially for prefrontal cortex and other evolutionarily expanded transmodal cortices,⁹⁸ it is likely that GABAergic signalling also played a role in the evolution of human transmodal association cortices. Intriguingly, although the majority of 5-HT_{2A}R-expressing cells are layer 5 pyramidal neurons, this receptor is also found on GABAergic interneurons in rodents, monkeys and humans,¹¹⁹ suggesting a noteworthy avenue for future research on their interactions for evolution and development. Thus, although our present account focuses on serotonin and 5-HT_{2A}R specifically, it should be understood in the context of the brain’s complex neuromodulatory landscape and the multiple influences on plasticity across spatial and temporal scales.

An integrative account of 5-HT_{2A}Rs in the development and adult function of human transmodal cortex

Taken together, there is considerable evidence indicating that human transmodal cortex exhibits a variety of unique structural and functional characteristics that collectively afford and underpin flexible, adaptive and complex aspects of behaviour and cognition. In addition, developmental and neuroanatomical evidence suggests strong linkages between 5-HT_{2A}R and the transmodal cortex, wherein this receptor may play a critical role in the expansion of such regions over development and allow for their potent functional-anatomical modulation in adulthood. Drawing together the various separate but converging lines of research presented in the previous sections, we propose an account of 5-HT_{2A}Rs as developmental drivers and adult modulators of the macroscale cortical processing hierarchy: 5-HT_{2A}Rs may play a critical role in facilitating the developmental expansion of the transmodal regions, which sit at the top of the hierarchy and then are well poised to potentially modulate its adult functioning when activated endogenously by

serotonin or exogenously by 5-HT_{2A}R agonist drugs. This account provides context for a deeper understanding of the therapeutic action of 5-HT_{2A}R agonist psychedelics when twinned with psychotherapeutic support.

5-HT_{2A}Rs as orchestrators of the cortical hierarchy

Thus, our account articulates (i) a developmental role for 5-HT_{2A}R in helping drive gyrencephalic cortical expansion in general and the disproportionate expansion of human transmodal cortex in particular; and (ii) a modulatory role for 5-HT_{2A}R in driving conditions for psychological change in the adult brain, via functional and neuroanatomical changes. As reviewed in the ‘5-HT_{2A} receptors as potent modulators and key developmental drivers of human transmodal cortex’ section, converging multimodal evidence indicates a critical role for 5-HT_{2A}Rs in stimulating the proliferation of basal progenitor cells, which are central to human cortical expansion.^{35,124,128,130} Moreover, research indicates that 5-HT_{2A}R densities in the adult human brain as measured *in vivo* are most expressed in regions of transmodal cortex, which underwent the greatest expansion in humans relative to phylogenetically proximal non-human primates¹⁸ (Fig. 1). These two independent sets of findings converge to suggest a process by which 5-HT_{2A}R agonism plays a causal role in transmodal cortical expansion during development and is subsequently positioned to modulate its functioning during adulthood. The cortical expansion engendered by 5-HT_{2A}R signalling in the early brain is mirrored by increased synaptic density in transmodal association cortices in the adult brain.^{95–97} Adult neuroplasticity is likely ideal for ongoing explorative learning—well-suited to complex, unpredictable environments.³²⁶ A modulatory role for 5-HT_{2A}R agonism over the activity and connectivity of transmodal cortex can be identified from functional MRI studies, with brain-wide consequences including an attenuation of the usual hierarchical differentiation of unimodal and transmodal cortex.^{166,229} This is reflected in behaviour as a potential dysregulation of top-down processing and increase in behavioural and cognitive flexibility.

From a functional and evolutionary perspective, in a non-drug context, one can intuit how a background of adversity and associated chronic stress e.g. conditions consistent with considerable evolutionary pressure, could prime a ‘growth’ or plasticity system (i.e. the 5-HT_{2A}R system) for engagement—in the service of environmental adaptation.¹⁷⁵ It is an evidence-informed speculation that this process is non-linear, i.e. upregulation of 5-HT_{2A}R reaches a ‘tipping’ or bifurcation point,¹⁷⁵ after which, with acute stress-induced release of 5-HT onto the primed 5-HT_{2A}R system, ideal (hyperplastic) conditions for a major state transition with potentially lasting sequelae, may ensue. When such triggering occurs (whether endogenously via stress-induced 5-HT release stress, or exogenously through 5-HT_{2A}R agonist psychedelics), increases in neuroplasticity and cognitive and psychological flexibility can occur, freeing dynamics from structural and top-down constraints and facilitating neural and cognitive exploration. The long-term effect of this process may be lasting psychological and behavioural change, where e.g. previously ‘stamped-in’ circuitry and associated psychological traits, can be made more plastic, i.e. amenable to change. If plasticity and learning are elevated for a prolonged period, as seems to be true with psychedelics,³¹ then the window for (re) learning (e.g. healthier traits) may endure well beyond the acute action of the drug.

As a consequence of uniquely human cortical expansion, the transmodal association cortex (where 5-HT_{2A}R expression is

greatest) moves farther away from the more ‘hard-coded’ unimodal cortices, becoming relatively less ‘tethered’ by molecular and structural constraints.⁶ Moreover, increased spatial and topographic distance of transmodal association cortex from unimodal cortices^{8,12} is accompanied by a corresponding reduction of functional-to-structural coupling,⁷³ and an increase of regional intrinsic timescale (i.e. longer temporal windows of integration).⁷⁵ Thus, as one progresses along the cortical hierarchy, regional activity becomes increasingly less determined by genetically encoded and structurally realized patterns of anatomy and connectivity, and also less determined by immediate sensorimotor contingencies—instead reflecting the higher-order transmodal, abstract integration of information across an extended period of time [i.e. tens of seconds instead of (milli)seconds].

Overall, we argue that the serotonin 2A receptor is involved in driving the expansion of the information-processing apex of the brain, i.e. the transmodal association cortex. In this context, there is recent evidence for consumption of 5-HT2AR-agonist mushrooms of the *Psilocybe* family by our hominin ancestors,³²⁷ pointing to the intriguing possibility of an active contribution to human brain evolution.²¹⁹ Moreover, after maturation to adulthood, 5-HT2ARs are uniquely poised to control this apex—and by implication—its governance of the rest of the brain (Fig. 5).

Challenges and future directions

Several questions naturally arise from the framework we have outlined in this article. For example, one may wonder about the case of prenatal exposure to selective serotonin reuptake inhibitors (SSRIs). By blocking SERT, SSRIs increase synaptic 5-HT levels and thereby increase signalling at 5-HT2ARs. Yet, there does not appear to be compelling evidence of an association between prenatal SSRI exposure and altered cortical development. Critically, however, this does not represent counterevidence to our proposal for two reasons. First, human NPCs do not express SERT during embryonic development, rendering SSRIs unable to increase 5-HT levels in the fetal brain.¹²⁸ Second, increased levels of maternal 5-HT induced by SSRIs cannot alter fetal progenitors, given that 5-HT does not pass the blood–brain barrier and therefore cannot reach the fetus from the mother’s brain.¹²⁷

Concordantly, the clinical literature on human prenatal exposure to 5-HT2A antagonists (including a variety of antipsychotic and antidepressant drugs, such as pimavanserin) has also not provided strong evidence of widespread alterations in cortical development. Indeed, based on the evidence from Xing and colleagues,³⁵ we predict that if neural progenitors in fetal human neocortex were directly exposed to high levels of a 5-HT2A antagonist, it would have neurodevelopmental repercussions on cortical volume. However, whether or not the antagonist can pass the placental barrier and reach the fetal brain is yet to be determined. Timing is also an important consideration: basal progenitors relevant to uniquely human cortical expansion are generated at approximately gestational week 10–16, such that out of this window, the effect of the antagonist on basal progenitors, hence on cortical development, should be rather limited. However, now that we have explicitly formulated our hypothesis, we hope that more specific investigations that take these factors (placental permeability and restricted temporal window) into consideration will be able to search for evidence of our proposed mechanism in humans. Such evidence could also be found in animals: our hypothesis predicts that reduced cortical volume should be found in 5-HT2AR knockout gyrencephalic animals where 5-HT2A is expressed in basal NPCs (e.g. ferret, pig or

non-human primates), but not lissencephalic mammals, such as mice, which do not exhibit 5-HT2A expression in basal NPCs. This is already suggested by the evidence of Xing and colleagues,³⁵ whereby disruption of 5-HT2AR expression in the ferret led to reduced levels of basal progenitors, especially proliferative basal progenitors—an effect that should result in a decrease in neurogenesis consequently reduced cortical volume. These observations also suggest that lissencephalic rodents may present some limitations as a model for evaluating possible side effects of novel drugs, given that they do not exhibit the key proliferation-inducing 5-HT2AR signalling role during development.

Another set of questions pertains to the regional distribution of the 2A receptor. For instance, although we have focused on the high availability of 2A receptors in transmodal association cortex, 5-HT2ARs are also densely expressed in primary visual cortex (and to a lesser extent, primary auditory cortex) of the human brain.^{18,112} Primary sensory cortices occupy the opposite end of the unimodal-transmodal hierarchy relative to transmodal cortex and it undergoes a more modest evolutionary expansion in humans relative to non-human primates.^{3,8} Primary cortex also does not undergo protracted development or plasticity, with its circuits predominantly defined in the first year of life following a brief critical period.²²² It is reasonable to wonder why this may be the case in light of the evidence we have reviewed about the role of 5-HT2ARs in promoting cortical expansion. We speculate that this may be in part attributable to the high level of intracortical myelination observed in V1,⁷ which is known to act as a physical and signalling barrier to plasticity after the end of cortical developmental maturation (which occurs much earlier in visual cortex relative to transmodal cortices).⁹³ This hypothesis is empirically testable and we hope that future research will take a closer look at potential non-neurotransmitter roles of 5-HT2ARs in V1. The ability of 5-HT2AR agonism to significantly increase functional connectivity between visual cortex and transmodal cortex is a consistent finding in functional MRI investigations^{161,163,166,223,229} and it is tempting to speculate that the potential for bridging the visual-transmodal gap may have developed in the service of complex and flexible behaviour. In this context, it is clear that further research is needed on the anatomical, functional and developmental differences between 5-HT2ARs in V1 and transmodal cortex.

On the other hand, there is evidence that the cerebellum has greatly expanded in evolutionary terms²²⁴—yet it displays little 5-HT2AR expression, despite receiving substantial serotonergic innervation targeting multiple 5-HT receptors.²²⁵ More broadly, we acknowledge that our cortico-centric account is inevitably incomplete, given that the subcortex and cerebellum play fundamental roles both in mental disorders and in healthy cognitive function³²⁹—with both structures displaying functional associations with cortical resting state networks, including transmodal cortex.³³⁰ Intriguingly, cerebellar functional gradients have also been identified including a principal sensory-fugal gradient.³³¹ Likewise, as the source of serotonergic and other neuromodulatory innervation to the brain, we anticipate that future extensions of the framework proposed here will feature a more prominent role for brainstem nuclei, such as the serotonergic raphe nuclei. Overall, we believe that a fuller account of how the dynamics and neuromodulation of complex, distributed systems can give rise to emergent high-level psychological phenomena,³³² is an important goal for future neuroscientific research.

Finally, we emphasize that the account of brain development and evolution provided here is not intended to be exhaustive. In addition to the abovementioned role of GABAergic signalling for

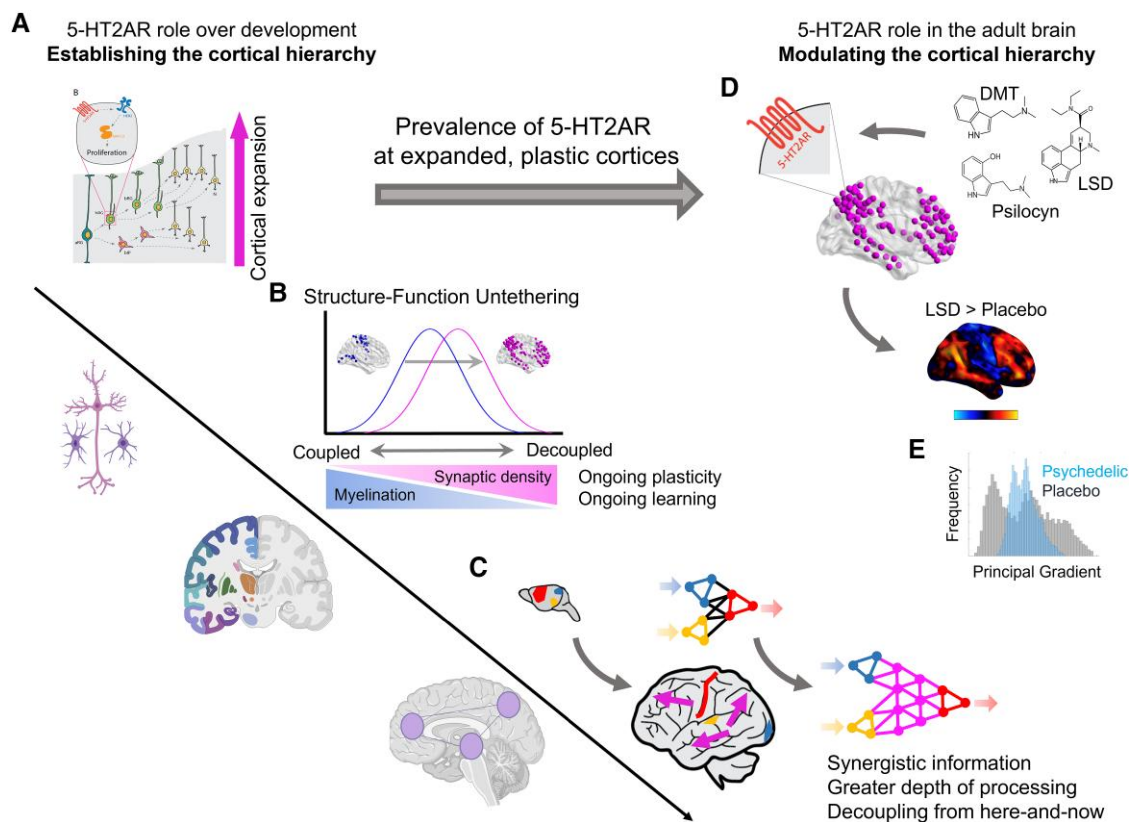


Figure 5 Schematic of the proposed dual roles of 5-HT_{2A}R in establishing (left) and then modulating (right) the human cortical hierarchy. (A–C) From the molecular to the cognitive level, 5-HT_{2A}Rs shape development and evolution by driving cortical expansion (A), inducing untethering of function from anatomical and genetic constraints, with greater synaptic density and lower intracortical myelination (B), and ultimately leading to a cognitive architecture with greater depth of processing thanks to the expansion of transmodal association cortex (C). (D and E) In the adult brain, 5-HT_{2A}R prevalence is elevated in transmodal association cortex and 5-HT_{2A}R engagement by serotonergic psychedelics (D) differentially affects the two ends of the cortical hierarchy, inducing a collapse of the principal functional gradient (E). Figure elements modified from Luppi et al.³²⁸ (under CC-BY license).

neural critical periods, it is clear that brain evolution can only be understood as a complex process, involving multiple mechanisms interacting across diverse temporal scales. Numerous genes have been implicated in development of the neocortex, whether due to the occurrence of microcephaly in animals and human patients upon their mutation³³³ or because of their role in stimulating neurogenesis and cortical expansion in the fetus.^{130,334,335} Additional biological factors include the role of differences in tissue oxygenation and metabolism^{48,94} and the concomitant availability of biosynthetic materials and plasticity-related genes in transmodal association cortices.⁹⁴ In turn, the metabolic burden of an expanded brain may have been paid for, at least in part, by the invention of cooking, an example of culturally-transmitted learning that increased the digestibility of food, thereby enabling more energy to be extracted in less time.^{336,337} In this context, it is intriguing to note that 95% of serotonin innervation is towards the gastrointestinal tract, where it plays a prominent role in digestion.¹⁷¹ Further supporting the role of culture and social interactions in brain evolution, the ability to learn from conspecifics³³⁸ and the need to outcompete members of one's own group^{339,340} and other groups³⁴¹ may have provided converging justifications for the advantage of neocortical expansion and the ensuing cognitive flexibility. Our account of the role of 5-HT_{2A}R adds another layer to this rich tapestry, towards understanding healthy and pathological brain function through the lens of development and evolution across scales.

Conclusion

In this multi-level synthesis, we have brought together human, non-human animal, *in vitro* and *in silico* evidence to show that serotonin 2A receptors are: (i) most densely expressed in transmodal association cortex—the apex of the human cortical hierarchy; (ii) play a key role in both the ontogenetic and phylogenetic development of the principal unimodal-transmodal hierarchical axis of the cortex; and (iii) have a unique ability to rapidly and potently modulate this hierarchy and the cognitive faculties and behaviours it encodes. By offering a unified account of the role of 5-HT_{2A}R in both the development and adult functioning of the human brain, this work stands to enrich the neurobiological and neuropharmacological understanding of human brain evolution. In turn, these insights will provide a crucial background for understanding the action of classic psychedelic drugs and we hope that they will inform ongoing research on the potential therapeutic applications of these compounds.

Funding

A.I.L. is supported by a Gates Cambridge Trust (OPP 1144). M.G. is supported by a Natural Sciences and Engineering Research Council of Canada CGS-D Doctoral Fellowship. E.A.S. is supported by the Stephen Erskine Fellowship of Queens' College, Cambridge. R.C.H. is supported by the Ralph Metzner Distinguished Professorship at UCSF.

Competing interests

M.G. reports receiving scientific advisory fees in the last 2 years from EntheoTech Bioscience. R.C.H. reports receiving scientific advisory fees in the last 2 years from: Beckley Psytech. All other authors report no conflicts of interest.

Supplementary material

Supplementary material is available at *Brain* online.

References

- Sydnor VJ, Larsen B, Bassett DS, et al. Neurodevelopment of the association cortices: Patterns, mechanisms, and implications for psychopathology. *Neuron*. 2021;109:2820–2846.
- Mesulam M. From sensation to cognition. *Brain*. 1998;121:1013–1052.
- Hill J, Inder T, Neil J, Dierker D, Harwell J, Van Essen D. Similar patterns of cortical expansion during human development and evolution. *Proc Natl Acad Sci U S A*. 2010;107:13135–13140.
- Wei Y, de Lange SC, Scholtens LH, et al. Genetic mapping and evolutionary analysis of human-expanded cognitive networks. *Nature Commun*. 2019;10:4839.
- Xu T, Nenning K-H, Schwartz E, et al. Cross-species functional alignment reveals evolutionary hierarchy within the connectome. *Neuroimage*. 2020;223:117346.
- Buckner RL, Krienen FM. The evolution of distributed association networks in the human brain. *Trends Cogn Sci*. 2013;17:648–665.
- Burt JB, Demirtaş M, Eckner WJ, et al. Hierarchy of transcriptional specialization across human cortex captured by structural neuroimaging topography. *Nat Neurosci*. 2018;21:1251–1259.
- Margulies DS, Ghosh SS, Goulas A, et al. Situating the default-mode network along a principal gradient of macroscale cortical organization. *Proc Natl Acad Sci U S A*. 2016;113:12574–12579.
- Hansen JY, Markello RD, Vogel JW, Seidlitz J, Bzdok D, Misic B. Mapping gene transcription and neurocognition across human neocortex. *Nat Hum Behav*. 2021;5:1240–1250.
- Goulas A, Changeux J-P, Wagstyl K, Amunts K, Palomero-Gallagher N, Hilgetag CC. The natural axis of transmitter receptor distribution in the human cerebral cortex. *Proc Natl Acad Sci U S A*. 2021;118:e2020574118.
- Valk SL, Xu T, Margulies DS, et al. Shaping brain structure: Genetic and phylogenetic axes of macroscale organization of cortical thickness. *Sci Adv*. 2020;6:eabb3417.
- Smallwood J, Bernhardt BC, Leech R, Bzdok D, Jefferies E, Margulies DS. The default mode network in cognition: A topographical perspective. *Nat Rev Neurosci*. 2021;22:503–513.
- Duncan J. The multiple-demand (MD) system of the primate brain: Mental programs for intelligent behaviour. *Trends Cogn Sci*. 2010;14:172–179.
- Menon V. Large-scale brain networks and psychopathology: A unifying triple network model. *Trends Cogn Sci*. 2011;15:483–506.
- Whitfield-Gabrieli S, Ford JM. Default mode network activity and connectivity in psychopathology. *Ann Rev Clin Psychol*. 2012;8:49–76.
- McClure-Begley TD, Roth BL. The promises and perils of psychedelic pharmacology for psychiatry. *Nat Rev Drug Discov*. 2022;21:1–2.
- Vollenweider FX, Preller KH. Psychedelic drugs: Neurobiology and potential for treatment of psychiatric disorders. *Nat Rev Neurosci*. 2020;21:611–624.
- Beliveau V, Ganz M, Feng L, et al. A high-resolution in vivo atlas of the human brain's serotonin system. *J Neurosci*. 2017;37:120–128.
- Johnson MW, Hendricks PS, Barrett FS, Griffiths RR. Classic psychedelics: An integrative review of epidemiology, therapeutics, mystical experience, and brain network function. *Pharmacol Ther*. 2019;197:83–102.
- Carhart-Harris R, Giribaldi B, Watts R, et al. Trial of psilocybin versus escitalopram for depression. *N Engl J Med*. 2021;384:1402–1411.
- Carhart-Harris RL, Bolstridge M, Rucker J, et al. Psilocybin with psychological support for treatment-resistant depression: An open-label feasibility study. *Lancet Psychiatry*. 2016;3:619–627.
- Davis AK, Barrett FS, May DG, et al. Effects of psilocybin-assisted therapy on major depressive disorder: A randomized clinical trial. *JAMA Psychiatry*. 2020;78:481–489.
- Palhano-Fontes F, Barreto D, Onias H, et al. Rapid antidepressant effects of the psychedelic ayahuasca in treatment-resistant depression: A randomized placebo-controlled trial. *Psychol Med*. 2019;49:655–663.
- Osório F, Sanches RF, Macedo LR, et al. Antidepressant effects of a single dose of ayahuasca in patients with recurrent depression: A preliminary report. *Braz J Psychiatry*. 2015;37:13–20.
- Ross S, Bossis A, Guss J, et al. Rapid and sustained symptom reduction following psilocybin treatment for anxiety and depression in patients with life-threatening cancer: A randomized controlled trial. *J Psychopharmacol*. 2016;30:1165–1180.
- Griffiths RR, Johnson MW, Carducci MA, et al. Psilocybin produces substantial and sustained decreases in depression and anxiety in patients with life-threatening cancer: A randomized double-blind trial. *J Psychopharmacol*. 2016;30:1181–1197.
- Johnson MW, Garcia-Romeu A, Cosimano MP, Griffiths RR. Pilot study of the 5-HT_{2A} agonist psilocybin in the treatment of tobacco addiction. *J Psychopharmacol*. 2014;28:983–992.
- Krebs TS, Johansen P-Ø. Lysergic acid diethylamide (LSD) for alcoholism: Meta-analysis of randomized controlled trials. *J Psychopharmacol*. 2012;26:994–1002.
- Ly C, Greb AC, Cameron LP, et al. Psychedelics promote structural and functional neural plasticity. *Cell Rep*. 2018;23:3170–3182.
- Raval NR, Johansen A, Donovan LL, et al. A single dose of psilocybin increases synaptic density and decreases 5-HT_{2A} receptor density in the pig brain. *Int J Mol Sci*. 2021;22:835.
- Shao L-X, Liao C, Gregg I, et al. Psilocybin induces rapid and persistent growth of dendritic spines in frontal cortex in vivo. *Neuron*. 2021;109:2535–2544.e4.
- Carhart-Harris RL, Friston KJ. REBUS And the anarchic brain: Toward a unified model of the brain action of psychedelics. *Pharmacol Rev*. 2019;71:316–344.
- Aleksandrova LR, Phillips AG. Neuroplasticity as a convergent mechanism of ketamine and classical psychedelics. *Trends Pharmacol Sci*. 2021;42:929–942.
- Vaid S, Huttner WB. Transcriptional regulators and human-specific/primate-specific genes in neocortical neurogenesis. *Int J Mol Sci*. 2020;21:4614.
- Xing L, Kalebic N, Namba T, Vaid S, Wimberger P, Huttner WB. Serotonin receptor 2A activation promotes evolutionarily relevant basal progenitor proliferation in the developing neocortex. *Neuron*. 2020;108:1113–1129.e6.
- Xing L, Huttner WB. Neurotransmitters as modulators of neural progenitor cell proliferation during mammalian neocortex development. *Front Cell Dev Biol*. 2020;8:391.

37. Gaspar P, Cases O, Maroteaux L. The developmental role of serotonin: News from mouse molecular genetics. *Nat Rev Neurosci.* 2003;4:1002-1012.
38. Brummelte S, Mc Glanaghy E, Bonnin A, Oberlander T. Developmental changes in serotonin signaling: Implications for early brain function, behavior and adaptation. *Neuroscience.* 2017;342:212-231.
39. Brodmann K. *Vergleichende lokalisationslehre der großhirnrinde in ihren prinzipien dargestellt auf grund des zellenbaues.* Barth; 1909.
40. Petrides M, Pandya D. Dorsolateral prefrontal cortex: Comparative cytoarchitectonic analysis in the human and the macaque brain and corticocortical connection patterns. *Eur J Neurosci.* 1999;11:1011-1036.
41. Felleman DJ, Van DE. Distributed hierarchical processing in the primate cerebral cortex. *Cereb Cortex.* 1991;1:1-47.
42. Sanides F. Comparative architectonics of the neocortex of mammals and their evolutionary interpretation. *Ann NY Acad Sci.* 1969;167:404-423.
43. Huntenburg JM, Bazin P-L, Margulies DS. Large-scale gradients in human cortical organization. *Trends Cogn Sci.* 2018;22:21-31.
44. Beul SF, Hilgetag CC. Neuron density fundamentally relates to architecture and connectivity of the primate cerebral cortex. *Neuroimage.* 2019;189:777-792.
45. Wagstyl K, Larocque S, Cucurull G, et al. Bigbrain 3D atlas of cortical layers: Cortical and laminar thickness gradients diverge in sensory and motor cortices. *PLoS Biol.* 2020;18: e3000678.
46. Paquola C, De Wael RV, Wagstyl K, et al. Microstructural and functional gradients are increasingly dissociated in transmodal cortices. *PLoS Biol.* 2019;17:e3000284.
47. Huntenburg JM, Bazin P-L, Goulas A, Tardif CL, Villringer A, Margulies DS. A systematic relationship between functional connectivity and intracortical myelin in the human cerebral cortex. *Cereb Cortex.* 2017;27:981-997.
48. Vaishnavi SN, Vlassenko AG, Rundle MM, Snyder AZ, Mintun MA, Raichle ME. Regional aerobic glycolysis in the human brain. *Proc Natl Acad Sci U S A.* 2010;107:17757-17762.
49. Elston GN, Fujita I. Pyramidal cell development: Postnatal spinyogenesis, dendritic growth, axon growth, and electrophysiology. *Front Neuroanat.* 2014;8:78.
50. Demirtaş M, Burt JB, Helmer M, et al. Hierarchical heterogeneity across human cortex shapes large-scale neural dynamics. *Neuron.* 2019;101:1181-1194.e13.
51. Wang X-J. Macroscopic gradients of synaptic excitation and inhibition in the neocortex. *Nat Rev Neurosci.* 2020;21:169-178.
52. Wagstyl K, Ronan L, Goodyer IM, Fletcher PC. Cortical thickness gradients in structural hierarchies. *Neuroimage.* 2015; 111:241-250.
53. la Fougère C, Grant S, Kostikov A, et al. Where in-vivo imaging meets cytoarchitectonics: The relationship between cortical thickness and neuronal density measured with high-resolution [18F] flumazenil-PET. *Neuroimage.* 2011;56:951-960.
54. Vézquez-Rodríguez B, Liu Z-Q, Hagmann P, Misic B. Signal propagation via cortical hierarchies. *Netw Neurosci.* 2020;4: 1072-1090.
55. Baum GL, Cui Z, Roalf DR, et al. Development of structure–function coupling in human brain networks during youth. *Proc Natl Acad Sci U S A.* 2020;117:771-778.
56. Preti MG, Van De Ville D. Decoupling of brain function from structure reveals regional behavioral specialization in humans. *Nat Commun.* 2019;10:4747.
57. Sepulcre J, Sabuncu MR, Yeo TB, Liu H, Johnson KA. Stepwise connectivity of the modal cortex reveals the multimodal organization of the human brain. *J Neurosci.* 2012;32:10649-10661.
58. Jones E, Powell T. An anatomical study of converging sensory pathways within the cerebral cortex of the monkey. *Brain.* 1970;93:793-820.
59. Goldman-Rakic PS. Topography of cognition: Parallel distributed networks in primate association cortex. *Ann Rev Neurosci.* 1988;11:137-156.
60. Fox KC, Shi L, Baek S, et al. Intrinsic network architecture predicts the effects elicited by intracranial electrical stimulation of the human brain. *Nat Hum Behav.* 2020;4: 1039-1052.
61. Murphy C, Jefferies E, Rueschemeyer S-A, et al. Distant from input: Evidence of regions within the default mode network supporting perceptually-decoupled and conceptually-guided cognition. *Neuroimage.* 2018;171:393-401.
62. Murphy C, Jefferies E, Rueschemeyer S-A, et al. Isolated from input: Transmodal cortex in the default mode network supports perceptually-decoupled and conceptually-guided cognition. *Neuroimage.* 2018;171:393-401.
63. Yeo BTT, Kirinenen FM, Sepulcre J, et al. The organization of the human cerebral cortex estimated by intrinsic functional connectivity. *J Neurophysiol.* 2011;106:1125-1165.
64. Smith SM, Fox PT, Miller KL, et al. Correspondence of the brain's functional architecture during activation and rest. *Proc Natl Acad Sci U S A.* 2009;106:13040-13045.
65. Niendam TA, Laird AR, Ray KL, Dean YM, Glahn DC, Carter CS. Meta-analytic evidence for a superordinate cognitive control network subserving diverse executive functions. *Cogn Affect Behav Neurosci.* 2012;12:241-268.
66. Buckner RL, DiNicola LM. The brain's default network: Updated anatomy, physiology and evolving insights. *Nat Rev Neurosci.* 2019;20:593-608.
67. Stawarczyk D, Bezdek MA, Zacks JM. Event representations and predictive processing: The role of the midline default network core. *Top Cogn Sci.* 2021;13:164-186.
68. Vatansever D, Menon DK, Stamatakis EA. Default mode contributions to automated information processing. *Proc Natl Acad Sci U S A.* 2017;114:12821-12826.
69. Margulies DS, Smallwood J. Converging evidence for the role of transmodal cortex in cognition. *Proc Natl Acad Sci U S A.* 2017; 114:12641-12643.
70. Spreng RN, DuPre E, Selarka D, et al. Goal-congruent default network activity facilitates cognitive control. *J Neurosci.* 2014; 34:14108-14114.
71. Andrews-Hanna JR, Smallwood J, Spreng RN. The default network and self-generated thought: Component processes and dynamic control. *Ann NY Acad Sci.* 2014;1316: 29-52.
72. Smallwood J, Turnbull A, Wang H-t, et al. The neural correlates of ongoing conscious thought. *iScience.* 2021;24:102132.
73. Vázquez-Rodríguez B, Suárez LE, Markello RD, et al. Gradients of structure–function tethering across neocortex. *Proc Natl Acad Sci U S A.* 2019;116:21219-21227.
74. Shafiei G, Markello RD, De Wael RV, Bernhardt BC, Fulcher BD, Misic B. Topographic gradients of intrinsic dynamics across neocortex. *Elife.* 2020;9:e62116.
75. Wolff A, Berberian N, Golesorkhi M, Gomez-Pilar J, Zilio F, Northoff G. Intrinsic neural timescales: Temporal integration and segregation. *Trends Cogn Sci.* 2022;26:159-173.
76. Cole MW, Reynolds JR, Power JD, Repovs G, Anticevic A, Braver TS. Multi-task connectivity reveals flexible hubs for adaptive task control. *Nat Neurosci.* 2013;16:1348-1355.
77. Yin D, Liu W, Zeljic K, et al. Dissociable changes of frontal and parietal cortices in inherent functional flexibility across the human life span. *J Neurosci.* 2016;36:10060-10074.

78. Yeo BT, Krienen FM, Eickhoff SB, et al. Functional specialization and flexibility in human association cortex. *Cereb Cortex*. 2015;25:3654–3672.
79. Burt JB, Preller KH, Demirtas M, et al. Transcriptomics-informed large-scale cortical model captures topography of pharmacological neuroimaging effects of LSD. *Elife*. 2021;10:e69320.
80. Valk SL, Xu T, Paquola C, et al. Genetic and phylogenetic uncoupling of structure and function in human transmodal cortex. *Nat Commun*. 2022;13:2341.
81. Kiebel SJ, Daunizeau J, Friston KJ. A hierarchy of time-scales and the brain. *PLoS Comput Biol*. 2008;4:e1000209.
82. Murray JD, Bernacchia A, Freedman DJ, et al. A hierarchy of intrinsic timescales across primate cortex. *Nat Neurosci*. 2014;17:1661–1663.
83. Honey CJ, Thesen T, Donner TH, et al. Slow cortical dynamics and the accumulation of information over long timescales. *Neuron*. 2012;76:423–434.
84. Hasson U, Yang E, Vallines I, Heeger DJ, Rubin N. A hierarchy of temporal receptive windows in human cortex. *J Neurosci*. 2008;28:2539–2550.
85. Baldassano C, Hasson U, Norman KA. Representation of real-world event schemas during narrative perception. *J Neurosci*. 2018;38:9689–9699.
86. Baldassano C, Chen J, Zadbood A, Pillow JW, Hasson U, Norman KA. Discovering event structure in continuous narrative perception and memory. *Neuron*. 2017;95:709–721.e5.
87. Gollo LL, Zalesky A, Hutchison RM, Van Den Heuvel M, Breakspear M. Dwelling quietly in the rich club: Brain network determinants of slow cortical fluctuations. *Philos Trans R Soc Lond B Biol Sci*. 2015;370:20140165.
88. Yeshurun Y, Nguyen M, Hasson U. The default mode network: Where the idiosyncratic self meets the shared social world. *Nat Rev Neurosci*. 2021;22:181–192.
89. Hofmann W, Schmeichel BJ, Baddeley AD. Executive functions and self-regulation. *Trends Cogn Sci*. 2012;16:174–180.
90. Uddin LQ. Cognitive and behavioural flexibility: Neural mechanisms and clinical considerations. *Nat Rev Neurosci*. 2021;22:167–179.
91. Braga RM, Leech R. Echoes of the brain: Local-scale representation of whole-brain functional networks within transmodal cortex. *Neuroscientist*. 2015;21:540–551.
92. Boghdadi AG, Teo L, Bourne JA. The involvement of the myelin-associated inhibitors and their receptors in CNS plasticity and injury. *Mol Neurobiol*. 2018;55:1831–1846.
93. Fletcher JL, Makowiecki K, Cullen CL, Young KM. Oligodendrogenesis and myelination regulate cortical development, plasticity and circuit function. *Semin Cell Dev Biol*. 2021;118:14–23.
94. Goyal MS, Hawrylycz M, Miller JA, Snyder AZ, Raichle ME. Aerobic glycolysis in the human brain is associated with development and neotenus gene expression. *Cell Metab*. 2014;19:49–57.
95. Jacobs B, Schall M, Prather M, et al. Regional dendritic and spine variation in human cerebral cortex: A quantitative Golgi study. *Cereb Cortex*. 2001;11:558–571.
96. Finnema SJ, Nabulsi NB, Eid T, et al. Imaging synaptic density in the living human brain. *Sci Transl Med*. 2016;8:348ra96.
97. Mak E, Holland N, Jones PS, et al. In vivo coupling of dendritic complexity with presynaptic density in primary tauopathies. *Neurobiol Aging*. 2021;101:187–198.
98. Gogtay N, Giedd JN, Lusk L, et al. Dynamic mapping of human cortical development during childhood through early adulthood. *Proc Natl Acad Sci U S A*. 2004;101:8174–8179.
99. Dong H-M, Margulies DS, Zuo X-N, Holmes AJ. Shifting gradients of macroscale cortical organization mark the transition from childhood to adolescence. *Proc Natl Acad Sci U S A*. 2021;118:e2024448118.
100. Liu X, Somel M, Tang L, et al. Extension of cortical synaptic development distinguishes humans from chimpanzees and macaques. *Genome Res*. 2012;22:611–622.
101. LeWinn KZ, Sheridan MA, Keyes KM, Hamilton A, McLaughlin KA. Sample composition alters associations between age and brain structure. *Nat Commun*. 2017;8:874.
102. Holtmaat A, Wilbrecht L, Knott GW, Welker E, Svoboda K. Experience-dependent and cell-type-specific spine growth in the neocortex. *Nature*. 2006;441:979–983.
103. Mueller S, Wang D, Fox MD, et al. Individual variability in functional connectivity architecture of the human brain. *Neuron*. 2013;77:586–595.
104. Hinton GE, Nowlan SJ. How learning can guide evolution. *Complex Syst*. 1987;1:495–502.
105. Byrne R, Whiten A. *Machiavellian intelligence: Social expertise and the evolution of intellect in monkeys, apes, and humans*. Oxford University Press; 1988.
106. Mars RB, Neubert F-X, Noonan MP, Sallet J, Toni I, Rushworth MF. On the relationship between the “default mode network” and the “social brain”. *Front Hum Neurosci*. 2012;6:189.
107. Donahue CJ, Glasser MF, Preuss TM, Rilling JK, Van Essen DC. Quantitative assessment of prefrontal cortex in humans relative to nonhuman primates. *Proc Natl Acad Sci U S A*. 2018;115:E5183–E5192.
108. Mediano PA, Rosas FE, Luppi AI, et al. Towards an extended taxonomy of information dynamics via integrated information decomposition. *arXiv*. [Preprint] arXiv:210913186.
109. Williams PL, Beer RD. Non-negative decomposition of multivariate information. *arXiv*. [Preprint] <https://doi.org/10.48550/arXiv.1004.2515>.
110. Luppi AI, Mediano PA, Rosas FE, et al. A synergistic core for human brain evolution and cognition. *Nat Neurosci*. 2022;25:771–782.
111. Ettrup A, Svarer C, McMahon B, et al. Serotonin 2A receptor agonist binding in the human brain with [11C] cimbi-36: Test-retest reproducibility and head-to-head comparison with the antagonist [18F] altanserine. *Neuroimage*. 2016;130:167–174.
112. López-Giménez JF, González-Maeso J. Hallucinogens and serotonin 5-HT 2A receptor-mediated signaling pathways. *Curr Top Behav Neurosci*. 2017;36:45–73.
113. Halberstadt AL, Geyer MA. Multiple receptors contribute to the behavioral effects of indoleamine hallucinogens. *Neuropharmacology*. 2011;61:364–381.
114. Markello RD, Misic B. Comparing spatial null models for brain maps. *Neuroimage*. 2021;236:118052.
115. Hall H, Farde L, Halldin C, Lundkvist C, Sedvall G. Autoradiographic localization of 5-HT2A receptors in the human brain using [3H] M100907 and [11C] M100907. *Synapse*. 2000;38:421–431.
116. Saulin A, Savli M, Lanzenberger R. Serotonin and molecular neuroimaging in humans using PET. *Amino Acids*. 2012;42:2039–2057.
117. Celada P, Puig M, Artigas F. Serotonin modulation of cortical neurons and networks. *Front Integr Neurosci*. 2013;7:25.
118. Jakab RL, Goldman-Rakic PS. 5-Hydroxytryptamine2A Serotonin receptors in the primate cerebral cortex: Possible site of action of hallucinogenic and antipsychotic drugs in pyramidal cell apical dendrites. *Proc Natl Acad Sci U S A*. 1998;95:735–740.

119. Willins DL, Deutch AY, Roth BL. Serotonin 5-HT2A receptors are expressed on pyramidal cells and interneurons in the rat cortex. *Synapse*. 1997;27:79–82.
120. Aru J, Suzuki M, Larkum ME. Cellular mechanisms of conscious processing. *Trends Cogn Sci*. 2020;24:814–825.
121. Ramaswamy S, Markram H. Anatomy and physiology of the thick-tufted layer 5 pyramidal neuron. *Front Cell Neurosci*. 2015;9:233.
122. Côté F, Fligny C, Bayard E, et al. Maternal serotonin is crucial for murine embryonic development. *Proc Natl Acad Sci U S A*. 2007;104:329–334.
123. Dooley AE, Pappas IS, Parnavelas JG. Serotonin promotes the survival of cortical glutamatergic neurons in vitro. *Exp Neurol*. 1997;148:205–214.
124. Mayer S, Chen J, Velmeshev D, et al. Multimodal single-cell analysis reveals physiological maturation in the developing human neocortex. *Neuron*. 2019;102:143–158.e7.
125. Toda T, Homma D, Tokuoka H, et al. Birth regulates the initiation of sensory map formation through serotonin signaling. *Dev Cell*. 2013;27:32–46.
126. Haydon P, McCobb D, Kater S. Serotonin selectively inhibits growth cone motility and synaptogenesis of specific identified neurons. *Science*. 1984;226:561–564.
127. Bonnin A, Levitt P. Fetal, maternal, and placental sources of serotonin and new implications for developmental programming of the brain. *Neuroscience*. 2011;197:1–7.
128. Pinson A, Huttner WB. Neocortex expansion in development and evolution—From genes to progenitor cell biology. *Curr Opin Cell Biol*. 2021;73:9–18.
129. Stepien BK, Vaid S, Huttner WB. Length of the neurogenic period—A key determinant for the generation of upper-layer neurons during neocortex development and evolution. *Front Cell Dev Biol*. 2021;9:676911.
130. Wilsch-Braeuninger M, Florio M, Huttner WB. Neocortex expansion in development and evolution—From cell biology to single genes. *Curr Opin Cell Biol*. 2016;39:122–132.
131. Lui JH, Hansen DV, Kriegstein AR. Development and evolution of the human neocortex. *Cell*. 2011;146:18–36.
132. Namba T, Huttner WB. Neural progenitor cells and their role in the development and evolutionary expansion of the neocortex. *Wiley Interdiscip Rev Dev Biol*. 2017;6:e256.
133. Noctor SC, Martínez-Cerdeño V, Ivic L, Kriegstein AR. Cortical neurons arise in symmetric and asymmetric division zones and migrate through specific phases. *Nat Neurosci*. 2004;7:136–144.
134. Hansen DV, Lui JH, Parker PR, Kriegstein AR. Neurogenic radial glia in the outer subventricular zone of human neocortex. *Nature*. 2010;464:554–561.
135. Fantegrossi WE, Gray BW, Bailey JM, Smith DA, Hansen M, Kristensen JL. Hallucinogen-like effects of 2-([2-(4-cyano-2,5-dimethoxyphenyl) ethylamino] methyl) phenol (25CN-NBOH), a novel N-benzylphenethylamine with 100-fold selectivity for 5-HT2A receptors, in mice. *Psychopharmacology (Berl)*. 2015;232:1039–1047.
136. Jensen AA, McCorvy JD, Leth-Petersen S, et al. Detailed characterization of the in vitro pharmacological and pharmacokinetic properties of N-(2-hydroxybenzyl)-2, 5-dimethoxy-4-cyanophenylethylamine (25CN-NBOH), a highly selective and brain-penetrant 5-HT2A receptor agonist. *J Pharmacol Exp Ther*. 2017;361:441–453.
137. Märcher Rørsted E, Jensen AA, Kristensen JL. 25CN-NBOH: A selective agonist for in vitro and in vivo investigations of the serotonin 2A receptor. *ChemMedChem*. 2021;16:3263–3270.
138. Stiles J, Jernigan TL. The basics of brain development. *Neuropsychol Rev*. 2010;20:327–348.
139. Fietz SA, Lachmann R, Brandl H, et al. Transcriptomes of germinal zones of human and mouse fetal neocortex suggest a role of extracellular matrix in progenitor self-renewal. *Proc Natl Acad Sci U S A*. 2012;109:11836–11841.
140. Stenzel D, Wilsch-Braeuninger M, Wong FK, Heuer H, Huttner WB. Integrin $\alpha\beta3$ and thyroid hormones promote expansion of progenitors in embryonic neocortex. *Development*. 2014;141:795–806.
141. Dehay C, Kennedy H. Cell-cycle control and cortical development. *Nat Rev Neurosci*. 2007;8:438–450.
142. Dehay C, Savatier P, Cortay V, Kennedy H. Cell-cycle kinetics of neocortical precursors are influenced by embryonic thalamic axons. *J Neurosci*. 2001;21:201–214.
143. Pinto L, Drechsel D, Schmid M-T, et al. AP2 γ Regulates basal progenitor fate in a region- and layer-specific manner in the developing cortex. *Nat Neurosci*. 2009;12:1229–1237.
144. Represa A, Ben-Ari Y. Trophic actions of GABA on neuronal development. *Trends Neurosci*. 2005;28:278–283.
145. LoTurco JJ, Owens DF, Heath MJ, Davis MB, Kriegstein AR. GABA And glutamate depolarize cortical progenitor cells and inhibit DNA synthesis. *Neuron*. 1995;15:1287–1298.
146. Haydar TF, Wang F, Schwartz ML, Rakic P. Differential modulation of proliferation in the neocortical ventricular and subventricular zones. *J Neurosci*. 2000;20:5764–5774.
147. Suzuki M, Nelson AD, Eickstaedt JB, Wallace K, Wright LS, Svendsen CN. Glutamate enhances proliferation and neurogenesis in human neural progenitor cell cultures derived from the fetal cortex. *Eur J Neurosci*. 2006;24:645–653.
148. Fukui M, Nakamichi N, Yoneyama M, et al. Up-regulation of ciliary neurotrophic factor receptor expression by GABAA receptors in undifferentiated neural progenitors of fetal mouse brain. *J Neurosci Res*. 2008;86:2615–2623.
149. Fukui M, Nakamichi N, Yoneyama M, et al. Modulation of cellular proliferation and differentiation through GABAB receptors expressed by undifferentiated neural progenitor cells isolated from fetal mouse brain. *J Cell Physiol*. 2008;216:507–519.
150. Garcia EE, Smith RL, Sanders-Bush E. Role of Gq protein in behavioral effects of the hallucinogenic drug 1-(2, 5-dimethoxy-4-iodophenyl)-2-aminopropane. *Neuropharmacology*. 2007;52:1671–1677.
151. Schmid CL, Raehal KM, Bohn LM. Agonist-directed signaling of the serotonin 2A receptor depends on β -arrestin-2 interactions in vivo. *Proc Natl Acad Sci U S A*. 2008;105:1079–1084.
152. Rodriguiz RM, Nadkarni V, Means CR, et al. LSD-stimulated behaviors in mice require β -arrestin 2 but not β -arrestin 1. *Sci Rep*. 2021;11:17690.
153. Aghajanian GK. Modeling “psychosis” in vitro by inducing disordered neuronal network activity in cortical brain slices. *Psychopharmacology (Berl)*. 2009;206:575–585.
154. Aghajanian GK, Marek GJ. Serotonin and hallucinogens. *Neuropsychopharmacology*. 1999;21:16S–23S.
155. Aghajanian G, Marek G. Serotonin induces excitatory postsynaptic potentials in apical dendrites of neocortical pyramidal cells. *Neuropharmacology*. 1997;36:589–599.
156. Jaster AM, González-Maeso J. Mechanisms and molecular targets surrounding the potential therapeutic effects of psychedelics. *Mol Psychiatry*. Published online 27 September 2023. doi:10.1038/s41380-023-02274-x
157. Vargas MV, Dunlap LE, Dong C, et al. Psychedelics promote neuroplasticity through the activation of intracellular 5-HT2A receptors. *Science*. 2023;379(6633):700–706.

158. Andrade R. Serotonergic regulation of neuronal excitability in the prefrontal cortex. *Neuropharmacology*. 2011;61:382–386.
159. Lambe EK, Aghajanian GK. Hallucinogen-induced UP states in the brain slice of rat prefrontal cortex: Role of glutamate spill-over and NR2B-NMDA receptors. *Neuropsychopharmacology*. 2006;31:1682–1689.
160. Timmermann C, Roseman L, Scharfner M, et al. Neural correlates of the DMT experience assessed with multivariate EEG. *Sci Rep*. 2019;9:1–13.
161. Carhart-Harris RL, Muthukumaraswamy S, Roseman L, et al. Neural correlates of the LSD experience revealed by multimodal neuroimaging. *Proc Natl Acad Sci U S A*. 2016;113:4853–4858.
162. Muthukumaraswamy SD, Carhart-Harris RL, Moran RJ, et al. Broadband cortical desynchronization underlies the human psychedelic state. *J Neurosci*. 2013;33:15171–15183.
163. Roseman L, Leech R, Feilding A, Nutt DJ, Carhart-Harris RL. The effects of psilocybin and MDMA on between-network resting state functional connectivity in healthy volunteers. *Front Hum Neurosci*. 2014;8:204.
164. Singleton SP, Luppi AI, Carhart-Harris RL, et al. LSD flattens the brain's energy landscape: evidence from receptor-informed network control theory. *bioRxiv*. [Preprint] <https://doi.org/10.1101/2021.05.14.444193>
165. Müller F, Dolder PC, Schmidt A, Liechti ME, Borgwardt S. Altered network hub connectivity after acute LSD administration. *Neuroimage Clin*. 2018;18:694–701.
166. Girn M, Roseman L, Bernhardt B, Smallwood J, Carhart-Harris R, Spreng RN. Serotonergic psychedelic drugs LSD and psilocybin reduce the hierarchical differentiation of unimodal and transmodal cortex. *Neuroimage*. 2022;256:119220.
167. Huang KW, Ochandarena NE, Philson AC, et al. Molecular and anatomical organization of the dorsal raphe nucleus. *Elife*. 2019;8:e46464.
168. Jacobs BL, Azmitia EC. Structure and function of the brain serotonin system. *Physiol Rev*. 1992;72:165–229.
169. Okaty BW, Commons KG, Dymecki SM. Embracing diversity in the 5-HT neuronal system. *Nat Rev Neurosci*. 2019;20:397–424.
170. Doya K, Miyazaki KW, Miyazaki K. Serotonergic modulation of cognitive computations. *Curr Opin Behav Sci*. 2021;38:116–123.
171. Shine JM, O'Callaghan C, Walpole IC, et al. Understanding the effects of serotonin in the brain through its role in the gastrointestinal tract. *Brain*. 2022;145:2967–2981.
172. Sharp T, Barnes NM. Central 5-HT receptors and their function; present and future. *Neuropharmacology*. 2020;177:108155.
173. Barnes NM, Sharp T. A review of central 5-HT receptors and their function. *Neuropharmacology*. 1999;38:1083–1152.
174. Carhart-Harris R, Nutt D. Serotonin and brain function: A tale of two receptors. *J Psychopharmacol*. 2017;31:1091–1120.
175. Brouwer A, Carhart-Harris RL. Pivotal mental states. *J Psychopharmacol*. 2021;35:319–352.
176. Murnane KS. Serotonin 2A receptors are a stress response system: Implications for post-traumatic stress disorder. *Behav Pharmacol*. 2019;30:151.
177. Epel ES. The geroscience agenda: Toxic stress, hormetic stress, and the rate of aging. *Ageing Res Rev*. 2020;63:101167.
178. McEwen BS. Protection and damage from acute and chronic stress: Allostasis and allostatic overload and relevance to the pathophysiology of psychiatric disorders. *Ann NY Acad Sci*. 2004;1032:1–7.
179. Dwivedi Y, Mondal AC, Payappagoudar GV, Rizavi HS. Differential regulation of serotonin (5HT) 2A receptor mRNA and protein levels after single and repeated stress in rat brain: Role in learned helplessness behavior. *Neuropharmacology*. 2005;48:204–214.
180. Fernandes C, McKittrick CR, File SE, McEwen BS. Decreased 5-HT_{1A} and increased 5-HT_{2A} receptor binding after chronic corticosterone associated with a behavioural indication of depression but not anxiety. *Psychoneuroendocrinology*. 1997;22:477–491.
181. Ossowska G, Nowak G, Kata R, Klenk-Majewska B, Danilczuk Z, Zebrowska-Lupina I. Brain monoamine receptors in a chronic unpredictable stress model in rats. *J Neural Transm (Vienna)*. 2001;108:311–319.
182. Takao K, Nagatani T, Kitamura Y, Kawasaki K, Hayakawa H, Yamawaki S. Chronic forced swim stress of rats increases frontal cortical 5-HT₂ receptors and the wet-dog shakes they mediate, but not frontal cortical β -adrenoceptors. *Eur J Pharmacol*. 1995;294:721–726.
183. Anju T, Smijin S, Korah P, Paulose C. Cortical 5HT_{2A} receptor function under hypoxia in neonatal rats: Role of glucose, oxygen, and epinephrine resuscitation. *J Mol Neurosci*. 2011;43:350–357.
184. Zhao X, Meyers K, McBride A, et al. Sleep deprivation rapidly upregulates serotonin 2A receptor expression via the immediate early gene Egr3. *bioRxiv*. [Preprint] <https://doi.org/10.1101/634410>
185. Maple AM, Zhao X, Elizalde DI, McBride AK, Gallitano AL. Htr2a expression responds rapidly to environmental stimuli in an Egr3-dependent manner. *ACS Chem Neurosci*. 2015;6:1137–1142.
186. Elmenhorst D, Kroll T, Matusch A, Bauer A. Sleep deprivation increases cerebral serotonin 2A receptor binding in humans. *Sleep*. 2012;35:1615–1623.
187. Cahir M, Ardis T, Reynolds GP, Cooper SJ. Acute and chronic tryptophan depletion differentially regulate central 5-HT_{1A} and 5-HT_{2A} receptor binding in the rat. *Psychopharmacology (Berl)*. 2007;190:497–506.
188. Hayashida S, Oka T, Mera T, Tsuji S. Repeated social defeat stress induces chronic hyperthermia in rats. *Physiol Behav*. 2010;101:124–131.
189. Preece M, Dalley J, Theobald D, Robbins T, Reynolds G. Region specific changes in forebrain 5-hydroxytryptamine_{1A} and 5-hydroxytryptamine_{2A} receptors in isolation-reared rats: An in vitro autoradiography study. *Neuroscience*. 2004;123:725–732.
190. Rilke O, Freier D, Jähkel M, Oehler J. Dynamic alterations of serotonergic metabolism and receptors during social isolation of low-and high-active mice. *Pharmacol Biochem Behav*. 1998;59:891–896.
191. Wright IK, Ismail H, Upton N, Marsden C. Effect of isolation rearing on 5-HT agonist-induced responses in the rat. *Psychopharmacology (Berl)*. 1991;105:259–263.
192. Benekareddy M, Goodfellow NM, Lambe EK, Vaidya VA. Enhanced function of prefrontal serotonin 5-HT₂ receptors in a rat model of psychiatric vulnerability. *J Neurosci*. 2010;30:12138–12150.
193. Sood A, Pati S, Bhattacharya A, Chaudhari K, Vaidya VA. Early emergence of altered 5-HT_{2A} receptor-evoked behavior, neural activation and gene expression following maternal separation. *Int J Dev Neurosci*. 2018;65:21–28.
194. Fujino K, Yoshitake T, Inoue O, et al. Increased serotonin release in mice frontal cortex and hippocampus induced by acute physiological stressors. *Neurosci Lett*. 2002;320:91–95.
195. Bastani A, Rajabi S, Kianimarkani F. The effects of fasting during ramadan on the concentration of serotonin, dopamine, brain-derived neurotrophic factor and nerve growth factor. *Neurol Int*. 2017;9:29–33.

196. Fuenmayor LD, García S. The effect of fasting on 5-hydroxytryptamine metabolism in brain regions of the albino rat. *Br J Pharmacol.* 1984;83:357–362.
197. Ishida A, Nakajima W, Takada G. Short-term fasting alters neonatal rat striatal dopamine levels and serotonin metabolism: An in vivo microdialysis study. *Dev Brain Res.* 1997;104:131–136.
198. Beitia G, Garmendia L, Azpiroz A, Vegas O, Brain P, Arregi A. Time-dependent behavioral, neurochemical, and immune consequences of repeated experiences of social defeat stress in male mice and the ameliorative effects of fluoxetine. *Brain Behav Immun.* 2005;19:530–539.
199. Gardner K, Thiruvikraman K, Lightman S, Plotsky P, Lowry C. Early life experience alters behavior during social defeat: Focus on serotonergic systems. *Neuroscience.* 2005;136:181–191.
200. Higuchi Y, Soga T, Parhar IS. Social defeat stress decreases mRNA for monoamine oxidase A and increases 5-HT turnover in the brain of male Nile tilapia (*Oreochromis niloticus*). *Front Pharmacol.* 2019;9:1549.
201. Keeney A, Jessop D, Harbuz M, Marsden C, Hogg S, Blackburn-Munro R. Differential effects of acute and chronic social defeat stress on hypothalamic-pituitary-adrenal axis function and hippocampal serotonin release in mice. *J Neuroendocrinol.* 2006;18:330–338.
202. Paul ED, Hale MW, Lukkes JL, Valentine MJ, Sarchet DM, Lowry CA. Repeated social defeat increases reactive emotional coping behavior and alters functional responses in serotonergic neurons in the rat dorsal raphe nucleus. *Physiol Behav.* 2011;104:272–282.
203. Neugebauer V. Serotonin—pain modulation. In: Müller CP, Cunningham KA, eds. *Handbook of behavioral neuroscience.* Vol 31. Elsevier; 2020:309–320.
204. Kupers R, Frokjaer VG, Naert A, et al. A PET [18F] altanserin study of 5-HT_{2A} receptor binding in the human brain and responses to painful heat stimulation. *Neuroimage.* 2009;44:1001–1007.
205. Brivio P, Sbrini G, Riva MA, Calabrese F. Acute stress induces cognitive improvement in the novel object recognition task by transiently modulating Bdnf in the prefrontal cortex of male rats. *Cell Mol Neurobiol.* 2020;40:1037–1047.
206. Vaidya VA, Marek GJ, Aghajanian GK, Duman RS. 5-HT_{2A} receptor-mediated regulation of brain-derived neurotrophic factor mRNA in the hippocampus and the neocortex. *J Neurosci.* 1997;17:2785–2795.
207. Nichols DE. Psychedelics. *Pharmacol Rev.* 2016;68:264–355.
208. Preller KH, Vollenweider FX. Phenomenology, structure, and dynamic of psychedelic states. *Curr Top Behav Neurosci.* 2016;36:221–256.
209. Studerus E, Gamma A, Vollenweider FX. Psychometric evaluation of the altered states of consciousness rating scale (OAV). *PLoS One.* 2010;5:e12412.
210. Andersen KA, Carhart-Harris R, Nutt DJ, Erritzoe D. Therapeutic effects of classic serotonergic psychedelics: A systematic review of modern-era clinical studies. *Acta Psychiatr Scand.* 2021;143:101–118.
211. Glennon RA, Titeler M, McKenney J. Evidence for 5-HT₂ involvement in the mechanism of action of hallucinogenic agents. *Life Sci.* 1984;35:2505–2511.
212. Canal CE, Morgan D. Head-twitch response in rodents induced by the hallucinogen 2, 5-dimethoxy-4-iodoamphetamine: A comprehensive history, a re-evaluation of mechanisms, and its utility as a model. *Drug Test Anal.* 2012;4:556–576.
213. Vollenweider FX, Vollenweider-Scherpenhuyzen MF, Bähler A, Vogel H, Hell D. Psilocybin induces schizophrenia-like psychosis in humans via a serotonin-2 agonist action. *Neuroreport.* 1998;9:3897–3902.
214. Stenbæk DS, Madsen MK, Ozenne B, et al. Brain serotonin 2A receptor binding predicts subjective temporal and mystical effects of psilocybin in healthy humans. *J Psychopharmacol.* 2021;35:459–468.
215. Preller KH, Burt JB, Ji JL, et al. Changes in global and thalamic brain connectivity in LSD-induced altered states of consciousness are attributable to the 5-HT_{2A} receptor. *Elife.* 2018;7:e35082.
216. Madsen MK, Fisher PM, Burmester D, et al. Psychedelic effects of psilocybin correlate with serotonin 2A receptor occupancy and plasma psilocin levels. *Neuropsychopharmacology.* 2019;44:1328–1334.
217. Desouza LA, Benekareddy M, Fanibunda SE, et al. The hallucinogenic serotonin(2A) receptor agonist, 2,5-dimethoxy-4-iodoamphetamine, promotes cAMP response element binding protein-dependent gene expression of specific plasticity-associated genes in the rodent neocortex. *Front Mol Neurosci.* 2021;14:790213.
218. Lombaert N, Hennes M, Gilissen S, et al. 5-HTR_{2A} And 5-HTR_{3A} but not 5-HTR_{1A} antagonism impairs the cross-modal reactivation of deprived visual cortex in adulthood. *Mol Brain.* 2018;11:65.
219. McKenna T. *Food of the gods: The search for the original tree of knowledge: A radical history of plants, drugs and human evolution.* Random House; 1999.
220. Kringelbach ML, Cruzat J, Cabral J, et al. Dynamic coupling of whole-brain neuronal and neurotransmitter systems. *Proc Natl Acad Sci U S A.* 2020;117:9566–9576.
221. Girn M, Rosas FE, Daws RE, Gallen CL, Gazzaley A, Carhart-Harris RL. A complex systems perspective on psychedelic brain action. *Trends Cogn Sci.* 2023;27:433–445.
222. Espinosa JS, Stryker MP. Development and plasticity of the primary visual cortex. *Neuron.* 2012;75:230–249.
223. Aqil M, Roseman L. More than meets the eye: The role of sensory dimensions in psychedelic brain dynamics, experience, and therapeutics. *Neuropharmacology.* 2022;223:109300.
224. Sereno MI, Diedrichsen J, Tachrount M, Testa-Silva G, d'Arceuil H, De Zeeuw C. The human cerebellum has almost 80% of the surface area of the neocortex. *Proc Natl Acad Sci U S A.* 2020;117:19538–19543.
225. Bishop GA, Ho R. The distribution and origin of serotonin immunoreactivity in the rat cerebellum. *Brain Res.* 1985;331:195–207.
226. Jones KA, Srivastava DP, Allen JA, Strachan RT, Roth BL, Penzes P. Rapid modulation of spine morphology by the 5-HT_{2A} serotonin receptor through kalirin-7 signaling. *Proc Natl Acad Sci U S A.* 2009;106:19575–19580.
227. Preller KH, Duerler P, Burt JB, et al. Psilocybin induces time-dependent changes in global functional connectivity: Psi-induced changes in brain connectivity. *Biol Psychiatry.* 2020;88:197–207.
228. Tagliazucchi E, Roseman L, Kaelen M, et al. Increased global functional connectivity correlates with LSD-induced ego dissolution. *Curr Biol.* 2016;26:1043–1050.
229. Timmermann C, Roseman L, Haridas S., et al. Human brain effects of DMT assessed via EEG-fMRI. *Proc Natl Acad Sci U S A.* 2023;120:e2218949120.
230. Gu S, Pasqualetti F, Cieslak M, et al. Controllability of structural brain networks. *Nat Commun.* 2015;6:8414.
231. Herzog R, Mediano PA, Rosas FE, et al. A mechanistic model of the neural entropy increase elicited by psychedelic drugs. *Sci Rep.* 2020;10:17725.

232. Deco G, Cruzat J, Cabral J, et al. Whole-brain multimodal neuroimaging model using serotonin receptor maps explains nonlinear functional effects of LSD. *Curr Biol*. 2018;28:3065-3074.e6.
233. Carhart-Harris RL. The entropic brain-revisited. *Neuropharmacology*. 2018;142:167-178.
234. Carhart-Harris RL, Leech R, Hellyer PJ, et al. The entropic brain: A theory of conscious states informed by neuroimaging research with psychedelic drugs. *Front Hum Neurosci*. 2014;8:20.
235. Schartner MM, Carhart-Harris RL, Barrett AB, Seth AK, Muthukumaraswamy SD. Increased spontaneous MEG signal diversity for psychoactive doses of ketamine, LSD and psilocybin. *Sci Rep*. 2017;7:46421.
236. Alamia A, Timmermann C, Nutt DJ, VanRullen R, Carhart-Harris RL. DMT Alters cortical travelling waves. *Elife*. 2020;9:e59784.
237. Pang 庞兆阳 Z, Alamia A, VanRullen R. Turning the stimulus on and off changes the direction of α traveling waves. *eNeuro*. 2020;7:ENEURO.0218-20.2020.
238. Toker D, Pappas I, Lendner JD, et al. Consciousness is supported by near-critical slow cortical electrodynamics. *Proc Natl Acad Sci U S A*. 2022;119:e2024455119.
239. Lebedev AV, Kaelen M, Lövdén M, et al. LSD-Induced entropic brain activity predicts subsequent personality change. *Hum Brain Mapp*. 2016;37:3203-3213.
240. Viol A, Palhano-Fontes F, Onias H, de Araujo DB, Hövel P, Viswanathan GM. Characterizing complex networks using entropy-degree diagrams: Unveiling changes in functional brain connectivity induced by ayahuasca. *Entropy*. 2019;21:128.
241. Viol A, Palhano-Fontes F, Onias H, de Araujo DB, Viswanathan G. Shannon entropy of brain functional complex networks under the influence of the psychedelic ayahuasca. *Sci Rep*. 2017;7:1-13.
242. Lord L-D, Expert P, Atasoy S, et al. Dynamical exploration of the repertoire of brain networks at rest is modulated by psilocybin. *Neuroimage*. 2019;199:127-142.
243. Atasoy S, Vohryzek J, Deco G, Carhart-Harris RL, Kringelbach ML. Common neural signatures of psychedelics: Frequency-specific energy changes and repertoire expansion revealed using connectome-harmonic decomposition. *Prog Brain Res*. 2018;242:97-120.
244. Atasoy S, Roseman L, Kaelen M, Kringelbach ML, Deco G, Carhart-Harris RL. Connectome-harmonic decomposition of human brain activity reveals dynamical repertoire reorganization under LSD. *Sci Rep*. 2017;7:17661.
245. Tagliazucchi E, Carhart-Harris R, Leech R, Nutt D, Chialvo DR. Enhanced repertoire of brain dynamical states during the psychedelic experience. *Hum Brain Mapp*. 2014;35:5442-5456.
246. Luppi AI, Carhart-Harris RL, Roseman L, Pappas I, Menon DK, Stamatakis EA. LSD Alters dynamic integration and segregation in the human brain. *Neuroimage*. 2021;227:117653.
247. Varley TF, Carhart-Harris R, Roseman L, Menon DK, Stamatakis EA. Serotonergic psychedelics LSD & psilocybin increase the fractal dimension of cortical brain activity in spatial and temporal domains. *Neuroimage*. 2020;220:117049.
248. Singleton SP, Luppi AI, Carhart-Harris RL, et al. Receptor-informed network control theory links LSD and psilocybin to a flattening of the brain's control energy landscape. *Nat Commun*. 2022;13:5812.
249. Alonso JF, Romero S, Mañanas MÀ, Riba J. Serotonergic psychedelics temporarily modify information transfer in humans. *Int J Neuropsychopharmacol*. 2015;18:pyv039.
250. Barnett L, Muthukumaraswamy SD, Carhart-Harris RL, Seth AK. Decreased directed functional connectivity in the psychedelic state. *Neuroimage*. 2020;209:116462.
251. Preller KH, Razi A, Zeidman P, Stämpfli P, Friston KJ, Vollenweider FX. Effective connectivity changes in LSD-induced altered states of consciousness in humans. *Proc Natl Acad Sci U S A*. 2019;116:2743-2748.
252. Friston KJ. Waves of prediction. *PLoS Biol*. 2019;17:e3000426.
253. Friston KJ. The free-energy principle: A unified brain theory? *Nat Rev Neurosci*. 2010;11:127-138.
254. Timmermann C, Kettner H, Letheby C, Roseman L, Rosas FE, Carhart-Harris RL. Psychedelics alter metaphysical beliefs. *Sci Rep*. 2021;11:22166.
255. Carhart-Harris RL. How do psychedelics work? *Curr Opin Psychiatry*. 2019;32:16-21.
256. Barlow RL, Alsö J, Jupp B, et al. Markers of serotonergic function in the orbitofrontal cortex and dorsal raphe nucleus predict individual variation in spatial-discrimination serial reversal learning. *Neuropsychopharmacology*. 2015;40:1619-1630.
257. Brigman JL, Mathur P, Harvey-White J, et al. Pharmacological or genetic inactivation of the serotonin transporter improves reversal learning in mice. *Cereb Cortex*. 2010;20:1955-1963.
258. Clarke H, Dalley J, Crofts H, Robbins T, Roberts A. Cognitive inflexibility after prefrontal serotonin depletion. *Science*. 2004;304:878-880.
259. Matias S, Lottem E, Dugué GP, Mainen ZF. Activity patterns of serotonin neurons underlying cognitive flexibility. *Elife*. 2017;6:e20552.
260. Rygula R, Clarke HF, Cardinal RN, et al. Role of central serotonin in anticipation of rewarding and punishing outcomes: Effects of selective amygdala or orbitofrontal 5-HT depletion. *Cereb Cortex*. 2015;25:3064-3076.
261. Walker S, Robbins T, Roberts A. Differential contributions of dopamine and serotonin to orbitofrontal cortex function in the marmoset. *Cereb Cortex*. 2009;19:889-898.
262. Romano AG, Quinn JL, Li L, et al. Intrahippocampal LSD accelerates learning and desensitizes the 5-HT2A receptor in the rabbit, Romano et al. *Psychopharmacology (Berl)*. 2010;212:441-448.
263. Harvey JA. Role of the serotonin 5-HT2A receptor in learning. *Learning & Memory*. 2003;10:355-362.
264. Kanen JW, Luo Q, Kandroodi MR, et al. Effect of lysergic acid diethylamide (LSD) on reinforcement learning in humans. *Psychol Med*. 2023;53:6434-6445.
265. Schultz W. Recent advances in understanding the role of phasic dopamine activity. *F1000Res*. 2019;8:1680.
266. Schultz W, Dayan P, Montague PR. A neural substrate of prediction and reward. *Science*. 1997;275:1593-1599.
267. Kuypers K, Riba J, De La Fuente Revenga M, Barker S, Theunissen E, Ramaekers J. Ayahuasca enhances creative divergent thinking while decreasing conventional convergent thinking. *Psychopharmacology (Berl)*. 2016;233:3395-3403.
268. Girn M, Mills C, Roseman L, Carhart-Harris RL, Christoff K. Updating the dynamic framework of thought: Creativity and psychedelics. *Neuroimage*. 2020;213:116726.
269. Family N, Vinson D, Vigliocco G, et al. Semantic activation in LSD: Evidence from picture naming. *Lang Cogn Neurosci*. 2016;31:1320-1327.
270. Wiefßner I, Falchi M, Maia LO, et al. LSD And creativity: Increased novelty and symbolic thinking, decreased utility and convergent thinking. *J Psychopharmacol*. 2022;36:348-359.
271. Doss MK, Považan M, Rosenberg MD, et al. Psilocybin therapy increases cognitive and neural flexibility in patients with major depressive disorder. *Transl Psychiatry*. 2021;11:574.
272. Mason N, Kuypers K, Reckweg J, et al. Spontaneous and deliberate creative cognition during and after psilocybin exposure. *Transl Psychiatry*. 2021;11:209.

273. Uthaug M, Van Oorsouw K, Kuypers K, et al. Sub-acute and long-term effects of ayahuasca on affect and cognitive thinking style and their association with ego dissolution. *Psychopharmacology (Berl)*. 2018;235:2979–2989.
274. Pokorny T, Duerler P, Seifritz E, Vollenweider FX, Preller KH. LSD Acutely impairs working memory, executive functions, and cognitive flexibility, but not risk-based decision-making. *Psychol Med*. 2020;50:2255–2264.
275. Murphy-Beiner A, Soar K. Ayahuasca's 'afterglow': Improved mindfulness and cognitive flexibility in ayahuasca drinkers. *Psychopharmacology (Berl)*. 2020;237:1161–1169.
276. Mason NL, Mischler E, Uthaug MV, Kuypers KP. Sub-acute effects of psilocybin on empathy, creative thinking, and subjective well-being. *J Psychoactive Drugs*. 2019;51:123–134.
277. MacLean KA, Johnson MW, Griffiths RR. Mystical experiences occasioned by the hallucinogen psilocybin lead to increases in the personality domain of openness. *J Psychopharmacol*. 2011;25:1453–1461.
278. Erritzoe D, Roseman L, Nour M, et al. Effects of psilocybin therapy on personality structure. *Acta Psychiatr Scand*. 2018;138:368–378.
279. Close JB, Hajien EC, Watts R, Roseman L, Carhart-Harris RL. Psychedelics and psychological flexibility—results of a prospective web-survey using the acceptance and action questionnaire II. *J Contextual Behav Sci*. 2020;16:37–44.
280. Davis AK, Barrett FS, Griffiths RR. Psychological flexibility mediates the relations between acute psychedelic effects and subjective decreases in depression and anxiety. *J Contextual Behav Sci*. 2020;15:39–45.
281. Goodkind M, Eickhoff SB, Oathes DJ, et al. Identification of a common neurobiological substrate for mental illness. *JAMA Psychiatry*. 2015;72:305–315.
282. Kaczurkin AN, Park SS, Sotiras A, et al. Evidence for dissociable linkage of dimensions of psychopathology to brain structure in youths. *Am J Psychiatry*. 2019;176:1000–1009.
283. Romer AL, Elliott ML, Knodt AR, et al. Pervasively thinner neocortex as a transdiagnostic feature of general psychopathology. *Am J Psychiatry*. 2021;178:174–182.
284. Consortium B, Anttila V, Bulik-Sullivan B, et al. Analysis of shared heritability in common disorders of the brain. *Science*. 2018;360:eaap8757.
285. Grasby KL, Jahanshad N, Painter JN, et al. The genetic architecture of the human cerebral cortex. *Science*. 2020;367:eaay6690.
286. Nutt D, Carhart-Harris R. The current Status of psychedelics in psychiatry. *JAMA Psychiatry*. 2021;78:121–122.
287. Moreno FA, Wiegand CB, Taitano EK, Delgado PL. Safety, tolerability, and efficacy of psilocybin in 9 patients with obsessive-compulsive disorder. *J Clin Psychiatry*. 2006;67:1735–1740.
288. Grob CS, Danforth AL, Chopra GS, et al. Pilot study of psilocybin treatment for anxiety in patients with advanced-stage cancer. *Arch Gen psychiatry*. 2011;68:71–78.
289. Gasser P, Kirchner K, Passie T. LSD-assisted psychotherapy for anxiety associated with a life-threatening disease: A qualitative study of acute and sustained subjective effects. *J Psychopharmacol*. 2014;29:57–68.
290. Bogenschutz MP, Forcehimes AA, Pommy JA, Wilcox CE, Barbosa P, Strassman RJ. Psilocybin-assisted treatment for alcohol dependence: A proof-of-concept study. *J Psychopharmacol*. 2015;29:289–299.
291. Gukasyan N, Davis AK, Barrett FS, et al. Efficacy and safety of psilocybin-assisted treatment for major depressive disorder: Prospective 12-month follow-up. *J Psychopharmacol*. 2022;36:151–158.
292. Anderson BT, Danforth A, Daroff R, et al. Psilocybin-assisted group therapy for demoralized older long-term AIDS survivor men: An open-label safety and feasibility pilot study. *EClinicalMedicine*. 2020;27:100538.
293. Goodwin GM, Aaronson ST, Alvarez O, et al. Single-dose psilocybin for a treatment-resistant episode of major depression. *N Engl J Med*. 2022;387:1637–1648.
294. von Rotz R, Schindowski EM, Jungwirth J, et al. Single-dose psilocybin-assisted therapy in major depressive disorder: A placebo-controlled, double-blind, randomised clinical trial. *EClinicalMedicine*. 2023;56:101809.
295. Holze F, Gasser P, Müller F, Dolder PC, Liechti ME. Lysergic acid diethylamide-assisted therapy in patients with anxiety with and without a life-threatening illness: A randomized, double-blind, placebo-controlled phase II study. *Biol Psychiatry*. 2023;93:215–223.
296. Schneider FR, Feusner J, Wheaton MG, et al. Pilot study of single-dose psilocybin for serotonin reuptake inhibitor-resistant body dysmorphic disorder. *J Psychiatr Res*. 2023;161:364–370.
297. Griffiths RR, Johnson MW, Richards WA, Richards BD, McCann U, Jesse R. Psilocybin occasioned mystical-type experiences: Immediate and persisting dose-related effects. *Psychopharmacology (Berl)*. 2011;218:649–665.
298. Griffiths RR, Richards WA, Johnson MW, McCann UD, Jesse R. Mystical-type experiences occasioned by psilocybin mediate the attribution of personal meaning and spiritual significance 14 months later. *J Psychopharmacol*. 2008;22:621–632.
299. Griffiths RR, Richards WA, McCann U, Jesse R. Psilocybin can occasion mystical-type experiences having substantial and sustained personal meaning and spiritual significance. *Psychopharmacology (Berl)*. 2006;187:268–283.
300. Schmid Y, Liechti ME. Long-lasting subjective effects of LSD in normal subjects. *Psychopharmacology (Berl)*. 2018;235:535–545.
301. Simonsson O, Sexton JD, Hendricks PS. Associations between lifetime classic psychedelic use and markers of physical health. *J Psychopharmacol*. 2021;35:447–452.
302. Hendricks PS, Thorne CB, Clark CB, Coombs DW, Johnson MW. Classic psychedelic use is associated with reduced psychological distress and suicidality in the United States adult population. *J Psychopharmacol*. 2015;29:280–288.
303. Ona G, Kohek M, Massaguer T, et al. Ayahuasca and public health: Health status, psychosocial well-being, lifestyle, and coping strategies in a large sample of ritual ayahuasca users. *J Psychoactive Drugs*. 2019;51:135–145.
304. Teixeira PJ, Johnson MW, Timmermann C, et al. Psychedelics and health behaviour change. *J Psychopharmacol*. 2022;36:12–19.
305. Mans K, Kettner H, Erritzoe D, Haijen EC, Kaelen M, Carhart-Harris RL. Sustained, multifaceted improvements in mental well-being following psychedelic experiences in a prospective opportunity sample. *Front Psychiatry*. 2021;12:647909.
306. Mitchell JM, Bogenschutz M, Lilienstein A, et al. MDMA-assisted therapy for severe PTSD: A randomized, double-blind, placebo-controlled phase 3 study. *Nat Med*. 2021;27:1025–1033.
307. Hartogsohn I. Set and setting, psychedelics and the placebo response: An extra-pharmacological perspective on psychopharmacology. *J Psychopharmacol*. 2016;30:1259–1267.
308. Carhart-Harris RL, Roseman L, Haijen E, et al. Psychedelics and the essential importance of context. *J Psychopharmacol*. 2018;32:725–731.
309. Vargas MV, Meyer R, Avanes AA, Rus M, Olson DE. Psychedelics and other psychoplastogens for treating mental illness. *Front Psychiatry*. 2021;12:1691.

310. De Vos CM, Mason NL, Kuypers KP. Psychedelics and neuroplasticity: A systematic review unraveling the biological underpinnings of psychedelics. *Front Psychiatry*. 2021;12:724606.
311. Branchi I. The double edged sword of neural plasticity: Increasing serotonin levels leads to both greater vulnerability to depression and improved capacity to recover. *Psychoneuroendocrinology*. 2011;36:339-351.
312. Lepow L, Morishita H, Yehuda R. Critical period plasticity as a framework for psychedelic-assisted psychotherapy. *Front Neurosci*. 2021;15:710004.
313. Nardou R, Lewis EM, Rothhaas R, et al. Oxytocin-dependent re-opening of a social reward learning critical period with MDMA. *Nature*. 2019;569:116-120.
314. Nardou R, Sawyer E, Song YJ, et al. Psychedelics reopen the social reward learning critical period. *Nature*. 2023;618:790-798.
315. Sloschower J, Guss J, Krause R, et al. Psilocybin-assisted therapy of major depressive disorder using acceptance and commitment therapy as a therapeutic frame. *J Contextual Behav Sci*. 2020;15:12-19.
316. Walsh Z, Thiessen MS. Psychedelics and the new behaviourism: Considering the integration of third-wave behaviour therapies with psychedelic-assisted therapy. *Int Rev Psychiatry*. 2018;30:343-349.
317. Watts R, Luoma JB. The use of the psychological flexibility model to support psychedelic assisted therapy. *J Contextual Behav Sci*. 2020;15:92-102.
318. Diering GH, Huganir RL. The AMPA receptor code of synaptic plasticity. *Neuron*. 2018;100:314-329.
319. Rao VR, Finkbeiner S. NMDA And AMPA receptors: Old channels, new tricks. *Trends Neurosci*. 2007;30:284-291.
320. Aghajanian GK, Marek GJ. Serotonin, via 5-HT_{2A} receptors, increases EPSCs in layer V pyramidal cells of prefrontal cortex by an asynchronous mode of glutamate release. *Brain Res*. 1999;825:161-171.
321. Palacios-Filardo J, Mellor JR. Neuromodulation of hippocampal long-term synaptic plasticity. *Curr Opin Neurobiol*. 2019;54:37-43.
322. Littman ML. Reinforcement learning improves behaviour from evaluative feedback. *Nature*. 2015;521:445-451.
323. Roelfsema PR, Holtmaat A. Control of synaptic plasticity in deep cortical networks. *Nat Rev Neurosci*. 2018;19:166-180.
324. Tang X, Jaenisch R, Sur M. The role of GABAergic signalling in neurodevelopmental disorders. *Nat Rev Neurosci*. 2021;22:290-307.
325. Hensch TK, Fagiolini M, Mataga N, Stryker MP, Baekkeskov S, Kash SF. Local GABA circuit control of experience-dependent plasticity in developing visual cortex. *Science*. 1998;282:1504-1508.
326. Waddington CH. Canalization of development and genetic assimilation of acquired characters. *Nature*. 1959;183:1654-1655.
327. Rodríguez Arce JM, Winkelman MJ. Psychedelics, sociality, and human evolution. *Front Psychol*. 2021;12:4333.
328. Luppi AI, Cain J, Spindler LR, et al. Mechanisms underlying disorders of consciousness: Bridging gaps to move toward an integrated translational science. *Neurocrit Care*. 2021;35:37-54.
329. Gellersen HM, Guo CC, O'Callaghan C, Tan RH, Sami S, Hornberger M. Cerebellar atrophy in neurodegeneration—A meta-analysis. *J Neurol Neurosurg Psychiatry*. 2017;88:780-788.
330. Li J, Curley WH, Guerin B, et al. Mapping the subcortical connectivity of the human default mode network. *Neuroimage*. 2021;245:118758.
331. Guell X, Schmähmann JD, Gabrieli JD, Ghosh SS. Functional gradients of the cerebellum. *Elife*. 2018;7:e36652.
332. Rosas FE, Mediano PA, Jensen HJ, et al. Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data. *PLoS Comput Biol*. 2020;16:e1008289.
333. Jayaraman D, Bae B-I, Walsh CA. The genetics of primary microcephaly. *Annu Rev Genomics Hum Genet*. 2018;19:177-200.
334. Kalebic N, Gilardi C, Stepien B, et al. Neocortical expansion due to increased proliferation of basal progenitors is linked to changes in their morphology. *Cell Stem Cell*. 2019;24:535-550.e9.
335. Florio M, Huttner WB. Neural progenitors, neurogenesis and the evolution of the neocortex. *Development*. 2014;141:2182-2194.
336. Aiello LC, Wheeler P. The expensive-tissue hypothesis: The brain and the digestive system in human and primate evolution. *Curr Anthropol*. 1995;36:199-221.
337. Huang CH, Yu X, Liao WB. The expensive-tissue hypothesis in vertebrates: Gut microbiota effect, a review. *Int J Mol Sci*. 2018;19:1792.
338. Molapour T, Hagan CC, Silston B, et al. Seven computations of the social brain. *Soc Cogn Affect Neurosci*. 2021;16:745-760.
339. Dunbar RI. The social brain hypothesis and its implications for social evolution. *Ann Hum Biol*. 2009;36:562-572.
340. Dunbar RIM. The social brain hypothesis. *Evol Anthropol*. 1998;6:178-190.
341. Ashton BJ, Kennedy P, Radford AN. Interactions with conspecific outsiders as drivers of cognitive evolution. *Nat Commun*. 2020;11:4937.